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Fiscal Policy in a Depressed Economy

ABSTRACT In a depressed economy, with short-term nominal interest rates at their zero lower bound, ample cyclical unemployment, and excess capacity, increased government purchases would be neither offset by the monetary authority raising interest rates nor neutralized by supply-side bottlenecks. Then even a small amount of hysteresis—even a small shadow cast on future potential output by the cyclical downturn—means, by simple arithmetic, that expansionary fiscal policy is likely to be self-financing. Even if it is not, it is highly likely to pass the sensible benefit-cost test of raising the present value of future potential output. Thus, at the zero bound, where the central bank cannot or will not but in any event does not perform its full role in stabilization policy, fiscal policy has the stabilization policy mission that others have convincingly argued it lacks in normal times. Whereas many economists have assumed that the path of potential output is invariant to even a deep and prolonged downturn, the available evidence raises a strong fear that hysteresis is indeed a factor. Although nothing in our analysis calls into question the importance of sustainable fiscal policies, it strongly suggests the need for caution regarding the pace of fiscal consolidation.

This paper examines fiscal policy in the context of an economy suffering, like the United States today, from protracted high unemployment and output short of potential. We argue that although the conventional wisdom articulated by John Taylor (2000) rejecting discretionary fiscal policy is appropriate in normal times, such policy has a major role to play in a severe downturn in the aftermath of a financial crisis that carries interest rates down to the zero nominal lower bound.

Our analysis reaches five conclusions about fiscal policy as a stabilization tool in a depressed as opposed to a normal economy:

—The absence of supply constraints in the short term, together with a binding zero lower bound on interest rates, means that the Keynesian

multiplier is likely to be substantially greater than the relatively small value it is thought to have in normal times. This multiplier may well be further magnified by an additional zero-bound effect: the impact of economic expansion on expected inflation and hence on real interest rates.

—At current and expected future real interest rates on government borrowing, even a very modest amount of “hysteresis,” through which cyclical output shortfalls affect the economy’s future potential, has a substantial effect on estimates of the impact of expansionary fiscal policy on the future debt burden. Although the data are far from conclusive, a number of fragments of evidence suggest that additional government spending that mitigates protracted output losses raises potential future output, even if the spending policies are not directly productive in themselves.¹

—Policies of austerity may well be counterproductive even by the yardstick of reducing the burden of financing the national debt in the future. Austerity in a depressed economy can erode the long-run fiscal balance. Stimulus can improve it.²

—Arguments that expansionary fiscal policy at the zero bound is not self-financing and does not pass a benefit-cost test by raising the present value of future potential output hinge on establishing one of three conditions: that monetary policy offsets the demand effects of fiscal policy even at the zero bound sufficiently that the multiplier is near zero, or that future potential output is invariant to the size and length of the downturn, or that interest rates are at or above the range seen historically, at least in the United States.

—Only when a government must pay a substantial premium over the social rate of time discount in order to borrow is the economy unlikely to benefit from expansionary fiscal policy at the zero bound.

The paper is organized as follows. Section I presents a highly stylized example making our basic point regarding self-financing fiscal policy. It then lays out an analytical framework for assessing the likelihood that expansionary fiscal policy will actually be expansionary, and it identifies the parameters that are most important in evaluating the impact of fiscal policy changes.

1. Of course, this case is strengthened and the long-term benefits of debt-financed government purchases at the zero bound are amplified if the government purchases themselves are directly productive and so boost the economy’s stock of public capital or private human capital.

2. This point was made a generation ago by Blanchard and Summers (1987). As Erceg and Linde (2010) recently put it, there could then be a “fiscal free lunch.”

Two further sections examine evidence on the central parameters in our framework: the fiscal multiplier and the extent of hysteresis. Both must be greater than zero for our central point to hold, yet both are subject to considerable uncertainty. Other key parameters are subject to less uncertainty: estimates of the expected future growth rate of potential output are tightly clustered; the financial market's estimate of real Treasury borrowing rates far into the future is public information.

Section II argues that the multiplier is context-dependent, depending in particular on the reaction function of monetary policy. It concludes that at moments like the present—when interest rates are constrained by the zero bound, the output gap is large, and cyclical unemployment is high—fiscal policy is likely to be more potent than standard estimates suggest. This conclusion boosts the benefits of expansionary fiscal policy in a depressed economy substantially, but, importantly, it does not depend on the policy-relevant multiplier being higher than standard estimates of the fiscal policy multiplier.

Section III examines the available evidence on the extent of hysteresis. Financial crises and demand-induced recessions appear to have an impact on potential output even after normal conditions are restored. This makes it plausible that measures that mitigate their effects would have long-run benefits. We find corroboration both in the behavior of economic forecasters and in a number of fragments of evidence on the effects of recessions.

Finally, section IV takes up issues relating to interest rates and monetary policy. It argues that available evidence on central bank behavior suggests that it is unlikely that, in a severely depressed economy, expansionary fiscal policy will lead to an offsetting monetary policy response. The section concludes with a discussion of policy implications of the analysis for the United States and the world. An appendix uses the framework laid out in section I to consider the conditions under which expansionary policy is not self-financing but nonetheless passes the benefit-cost test of raising the present value of output—what we call the “extra-output benefit-cost test.”

I. Self-Financing Fiscal Policy

Assume an economy in which output is well below its potential, cyclical unemployment is elevated, supply constraints on short-run demand are absent, conventional monetary policy is constrained by the zero lower bound, and the central bank is either unable or unwilling to, but in any

case does not, provide additional stimulus through quantitative easing or other means (an assumption we discuss further in section IV).³ A simple calculation then conveys the main message of this paper: under these circumstances, a combination of real government borrowing rates in the historical range, modestly positive fiscal multiplier effects, and small hysteresis effects are together sufficient to render fiscal expansion self-financing.

Imagine, for example, that in this demand-constrained economy the fiscal multiplier is 1.5, the real annual interest rate on long-term government debt is 1 percent, a \$1 increase in GDP increases the net tax-and-transfer fiscal balance by \$0.33, and a \$1 shortfall of GDP below potential this year permanently reduces future potential GDP by \$0.01—that is, a hysteresis “shadow” on future potential output of only 1 percent. Assume further that the government has the power to undertake a transitory increase in spending and then reverse it without any impact on the risk premium that it pays on its borrowing.

Under these assumptions, the effect of an incremental \$1 of government spending is to increase current GDP by \$1.50 and to raise the debt by \$0.50. The annual real debt service on this additional debt is \$0.005. The \$1.50 increase in this year’s GDP increases future potential output by \$0.015, which in turn augments future-period tax revenue by \$0.005, on the assumption that actual output averages to potential output over the relevant future periods. Hence the fiscal expansion is self-financing. In such a scenario, worries about the adverse impact of fiscal stimulus on the government’s long-run budget are unwarranted, for there is no adverse impact.

This central point is made substantially stronger if one allows for:

- underlying growth in the economy, so that the relevant fiscal balance requirement is one of a stable debt-to-GDP ratio rather than a stable real debt;

- increases in the future price level, as a result of the fiscal expansion, that further reduce the real interest rate on accumulated and newly issued debt; and

3. Most estimates of Federal Reserve reaction functions suggest that, if it were possible to have negative short-term safe nominal interest rates, such rates would have been chosen in recent years. This fact indicates the relevance of our analysis. See Rudebusch (2009) and Taylor (2010).

—the possibility that the additional government spending raises future productivity, and thus future output, by increasing the productive stocks of public infrastructure capital and private human capital.⁴

This central point is a matter of arithmetic. It depends only on the existence of a fiscal multiplier μ that is not near zero, the existence of a plausible hysteresis shadow on future potential output, low and unchanged government borrowing costs, and the assumption that a temporary boost to government purchases is possible. If these four assumptions are granted, the conclusion follows.

This section presents a reduced-form framework for assessing under what conditions fiscal expansion is self-financing; the appendix discusses the conditions under which, if fiscal expansion is not self-financing, it nonetheless passes an extra-output benefit-cost test. Our conclusions will apply to any underlying model that generates such a reduced form.

A temporary boost to government purchases ΔG boosts aggregate demand through the short-term fiscal multiplier. More formally, an increase in government spending for the present period only of ΔG percentage-point-of-potential-GDP-years is amplified by the economy's short-term policy-relevant multiplier coefficient μ , reducing the output gap in the present period Y_n ("n" for "now") by an amount ΔY_n , also measured in percentage-point-years:

$$(1) \quad \Delta Y_n = \mu \Delta G.$$

We discuss in section II the value of μ in normal times and make the crucial point that there is a strong likelihood that μ is now above that value.

Financing this expansion of government purchases requires increasing the national debt by an amount ΔD , also measured in percentage-point-of-potential-GDP-years. Given μ as before and assuming a baseline marginal tax-and-transfer rate τ , the required increase in the national debt is then

$$(2) \quad \Delta D = (1 - \mu\tau) \Delta G,$$

4. It is worth stressing that with current real Treasury interest rates near zero (some estimates are provided later in this section), even if additional spending had no impact on current GDP, every government investment project that promises a positive real rate of return of any magnitude would boost the present value of future real GDP.

which is less than in the absence of the multiplier because higher current output brings with it higher tax collections and thus an immediate partial recapture of some of the costs of the fiscal expansion.

If the economy's long-run growth rate is g and the real government borrowing rate is r ,⁵ this additional debt ΔD imposes on the government an annual financing burden in percentage points of a year's potential GDP of

$$(3) \quad (r - g)\Delta D = (r - g)(1 - \mu\tau)\Delta G,$$

if it is to maintain a stable long-run debt-to-GDP ratio. In order to maintain a stable debt-to-GDP ratio, the government must increase its primary surplus by the difference between the growth rates of the debt and of GDP times the increment to the debt. That is the left-hand side of equation 3. And the increment to the debt is simply $(1 - \mu\tau)\Delta G$.

A depressed economy is one in which many workers are without employment for an extended period. As a consequence, many see their skills, the networks they use to match themselves with vacancies in the labor market, and their morale all decay. A depressed economy is also one in which investment is low, the capital stock is growing slowly if at all, and entrepreneurial exploration is low, and it is certainly possible that this deficit is not made up quickly. These factors may well have an impact on future potential output.

Assume that in future periods production is determined by supply and that there is no gap between real aggregate demand and potential output. Then, in a typical future period, potential and actual output Y_f (where “ f ” stands for “future”) will be reduced by a hysteresis parameter η times the depth by which the economy is depressed in the present:

$$(4) \quad \Delta Y_f = \eta \Delta Y_n = \eta \mu \Delta G.$$

The units of η are inverse years: η is the percent reduction in the flow of future potential output per percentage-point-year of the present-period output gap. We discuss the mechanisms determining η in section III.

A fiscal expansion undertaken to prevent hysteresis thus creates a fiscal dividend: it raises future tax collections by an amount

$$(5) \quad \tau \Delta Y_f = \tau \eta \mu \Delta G.$$

5. In the main text of this paper, r refers to both the social rate of time discount and the government's borrowing rate. The appendix considers the case when these two need to be distinguished.

Equations 3 and 5 together imply that if

$$(6) \quad (r - g)(1 - \mu\tau) - \eta\mu\tau \leq 0,$$

then at the margin, transitory expansionary fiscal policy is self-financing. The boost to future potential output, and thus to future net tax revenue, provided by shortening and lessening the current downturn creates more public financial resources in the future than are consumed by amortizing the additional debt incurred to finance the transitory expansion. There is no cost to count against this benefit from future fiscal expansion. This is the most important conclusion of this paper.

Rearranging equation 6, we can show that this net future fiscal dividend from the present-period fiscal expansion ΔG arises as long as r satisfies

$$(7) \quad r \leq g + \frac{\eta\mu\tau}{(1 - \mu\tau)}.$$

As long as there is a short-term fiscal multiplier μ , a hysteresis shadow η , a tax-and-transfer share τ , a real government borrowing rate r , and a debt amortization equation incorporating a trend growth rate g such that expression 7 holds, fiscal expansion now improves the government's budget balance later.⁶ In this case, arguments that a depressed economy cannot afford fiscal expansion now because the government dare not raise its debt have little purchase.⁷ And arguments that governments in such circumstances need to demonstrate the credibility of their long-run fiscal strategy by curbing spending today lack coherence, because cutting spending does not improve but rather worsens the long-run fiscal picture.

For what values in the parameter space does expression 7 hold, if we take the marginal tax rate τ and the expected rate of long-run GDP growth g to be their consensus values? For the marginal net tax-and-transfer share τ , we assume the baseline value to be 0.333. For g , the long-term growth rate of real potential GDP, we take the Congressional Budget Office's current estimate of 2.5 percent per year. This leaves μ and η —the fiscal multiplier and the hysteresis coefficient that captures the shadow cast by the downturn on the long run—as variable parameters. We take the plausible range for μ

6. For a somewhat different argument that austerity worsens the government's budget balance, see Denes, Eggertsson, and Gilbukh (2012).

7. This point is by no means new: see Lerner (1943). Wray (2002) argues that Milton Friedman's post-World War II proposal for stabilization policy achieved through a money supply provided by countercyclical deficit financing and 100 percent reserve banking is in its essence the same idea.

Table 1. Parameter Values for the Base Case

<i>Parameter</i>	<i>Interpretation</i>	<i>Assumed value</i>
μ	Present-period government spending multiplier	0–2.5
r	Real government borrowing rate and social rate of time discount, per year	0.025–?
g	Trend growth rate of potential GDP, per year	0.025
τ	Marginal tax-and-transfer rate	0.333
ξ	Disincentive effect: reduction in potential output from raising additional tax revenue	0.25–0.5
η	Hysteresis effect: proportional reduction in potential output from a temporary downturn	0–0.2

in a severely depressed economy at the zero lower bound to be between zero and 2.5, and the plausible range for η to be between zero and 0.2. Table 1 summarizes the framework parameters and their base-case values.

When calibrating η , it is probably best to consider it as a “permanent equivalent” concept. Short-term Keynesian effects die out in less than 5 years; permanent effects are forever. In a growing economy, permanent effects are thus capitalized at a multiple of $1/(r - g)$, which for plausible borrowing rates and social rates of time discount r , and plausible growth rates g , can be a very large factor. However, many plausible channels through which a deep and prolonged downturn casts a shadow on future potential output produce not permanent but rather persistent effects: they last for one generation, but not for three.

We therefore consider η to be the size of the persistent effects of a downturn on potential output in a permanent-equivalent metric: that is, we correct for the fact that these effects are long-run but not truly permanent, and hence should be capitalized not at a factor $1/(r - g)$ but rather at $[1 - (1 - r + g)^T]/(r - g)$, where T captures the length of the persistent but not truly permanent effects.

Table 2 reports critical Treasury borrowing rates below which expansionary fiscal policy is self-financing (expression 7 holds) for various values of η and μ . For example, for a multiplier μ of 1.5 and a hysteresis parameter η of 0.10, the second term on the right-hand side of expression 7 is 10 percent per year. This means that if the spread between the real Treasury borrowing rate r and the real growth rate of GDP g is less than 10 percentage points per year, fiscal expansion today improves rather than degrades the long-term budget balance of the government. Given our assumption that $g = 2.5$ percent, that implies a real Treasury borrowing rate of 17.5 percent per year or less.

Table 2. Critical Values of the Real Treasury Rate for Fiscal Expansion to Be Self-Financing

<i>Hysteresis</i> η	<i>Critical real Treasury interest rate for indicated value of multiplier μ (percent per year)^a</i>				
	$\mu = 0$	$\mu = 0.5$	$\mu = 1.0$	$\mu = 1.5$	$\mu = 2.5$
0	2.50	2.50	2.50	2.50	2.50
0.025	2.50	2.99	3.73	4.95	14.29
0.050	2.50	3.49	4.96	7.40	26.07
0.100	2.50	4.48	7.43	12.30	49.64
0.200	2.50	6.45	12.35	22.10	96.97

Source: Authors' calculations.

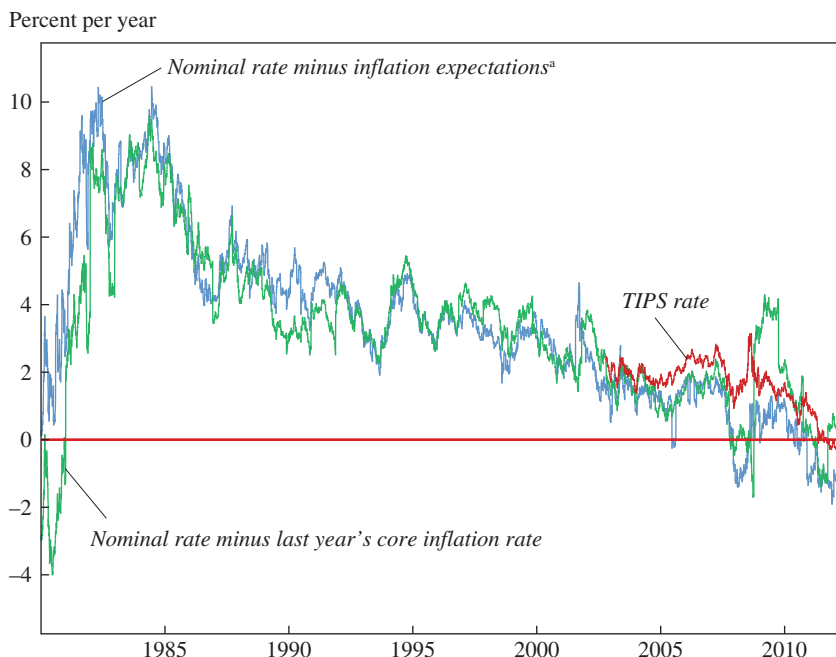
a. The critical rate is the highest rate that satisfies expression 7 in the text. Other parameters take the values assumed in table 1.

For μ of 1.0 and η of 0.1, the second term on the right-hand side of expression 7 is about 5 percent per year. In this case, if the spread between r and g is less than about 5 percentage points, fiscal expansion today improves rather than degrades the long-term budget balance of the government. That implies a real Treasury borrowing rate of about 7.5 percent per year or less.

For μ of 0.5 and η of 0.05, the second term on the right in expression 7 is about 1 percent per year. In this case, if the spread between r and g is less than about 1 percentage point, fiscal expansion today improves rather than degrades the long-term budget balance of the government. That implies a real Treasury borrowing rate of about 3.5 percent per year or less.

How credible is the claim that the Treasury's borrowing rates will stay below the relevant value in table 2, and thus that expansionary fiscal policy would be self-financing? Since January 1997 the interest rates on Treasury inflation-protected securities (TIPS) provide a direct, market-based measure of the real rate at which the Treasury can borrow. For earlier periods, subtracting a measure of the inflation rate from nominal interest rates provides a proxy. Figure 1 plots, in addition to the yield on 10-year TIPS, two such proxies: the yield on 10-year nominal Treasuries minus expected inflation from the University of Michigan Survey, and the same 10-year nominal yield minus the previous year's core inflation rate. These two measures do not markedly or persistently diverge from the TIPS rate over the period for which the latter is available. The expectations-based measure shows a somewhat higher mean value and more variability, but since the Volcker disinflation of the early 1980s it has tracked or undershot the current value of inflation.

The multiplier μ has to be low and the hysteresis parameter η almost negligible for the critical interest rate r to lie above the range of real interest

Figure 1. Proxies for Real Ten-Year Treasury Interest Rates, 1980–2012

Sources: FRED database, Federal Reserve Bank of St. Louis; University of Michigan; Bureau of Labor Statistics.

a. Inflation expectations are measured as the median expected percent change in prices over the next 12 months from the Thomson Reuters/University of Michigan Surveys of Consumers.

rates on Treasury debt seen in the historical record. At a real interest rate of 5 percent per year, expansionary fiscal policy is self-financing for $\mu = 2.5$ as long as $\eta > 0.005$; it is self-financing for $\mu = 1.5$ as long as $\eta > 0.025$; it is self-financing for $\mu = 1.0$ as long as $\eta > 0.050$; and it is self-financing for $\mu = 0.5$ as long as $\eta > 0.125$. The case for expansionary U.S. fiscal policy imposing any significant budgetary cost thus appears to rest on a claim that μ is significantly less than 1.0, or that η is significantly less than 0.05.

Moreover, current and expected future interest rates today are much lower than in the historical post-World War II experience, and today's long-term Treasury rates indicate that r is expected to stay extraordinarily low for a generation. As of June 1, 2012, the 10- and 30-year nominal Treasury rates were 1.47 and 2.53 percent per year, respectively; the 10- and 30-year TIPS rates were -0.59 and $+0.36$ percent per year, respectively—and many market observers see TIPS rates as elevated today because of perceived

lack of liquidity. If there is no expected term premium—if the expectations theory of the term structure holds—then financial markets currently anticipate that the short-term nominal Treasury rate will average less than 1.47 percent per year over the next 10 years, and less than 2.53 percent per year over the next 30 years. These are extraordinarily low rates. At an expected annual inflation rate of 2.0 percent and an expected real annual GDP growth rate of 2.5 percent, 1 percent of GDP worth of debt borrowed now and funded for 30 years with no nominal amortization raises the debt-to-GDP ratio a generation hence by only 0.55 percentage point. Assuming log utility and a zero rate of pure time preference, public spending that has a current-dollar benefit-cost ratio of only 0.55 is worth undertaking today as long as it can be funded with 30-year Treasuries.

Moreover, it is extremely unlikely that the expectations theory of the term structure holds without any term premium driving a wedge between expected future short-term rates and the current 30-year Treasury bond rate. If the past generation's detailed investigations into financial markets have taught us anything, it is that a great many risks that do not have a clear correlation with the marginal utility of aggregate consumption are nevertheless priced, indeed priced substantially. The risk that the value of one's long-term bonds will be eroded by inflation has been priced in the past through a considerable term premium relative to the expectations hypothesis of the term structure. It is hard to see any reason for this historical correlation to fail to hold in the future. This means that the arithmetic of government spending now is even more favorable, for markets do not anticipate a return of interest rates to their postwar norm for at least a generation.

At this point a very natural question arises: if interest rates on Treasury debt are usually (except in the early 1980s) sufficiently low to allow the government to borrow, spend, and end up with no net increase in its debt burden, why not do so always? The principal reason is that it cannot do so in normal times. A multiplier μ of even 1 is, as we discuss in section II, likely to be unusual. It is likely to prevail only when the zero lower bound on short-term interest rates is binding and cyclical unemployment is substantial. At other, normal times, μ is likely to be much smaller than 1. When interest rates are away from their zero bound, when the output gap is small, or when high unemployment is not cyclical but structural, then either bottlenecks or monetary policy offset make it unlikely that fiscal expansion will impart any significant boost to real GDP. When that is so, there is no stabilization policy case for expansionary fiscal policy.

Note that the arithmetic of table 2 does not hinge on the economy being close to the edge of or in the range of dynamic inefficiency. The key interest

rate in table 2 is r , and here it matters that r is the real interest rate on government borrowing and not the private marginal product of capital, the real social rate of time discount, or the rate of return on public capital.⁸

The conclusion that fiscal expansion may be self-financing is at least partially a point about the attractiveness of Treasury debt to investors (see Krishnamurthy and Vissing-Jorgensen 2012). If government debt is sufficiently attractive as a safe savings vehicle, and if there are at least minor counterhysteresis benefits from expansionary fiscal policy, then there need be no net financing burden of extra government purchases on taxpayers. Thus, the government can borrow, spend to boost the economy, use the extra taxes from a more prosperous economy to amortize part of its debt, refinance the debt and so push out the time horizon at which it is to be retired, and, as that horizon is extended, watch the debt-to-GDP ratio fall indefinitely. This would not be possible if Treasury debt were unattractive, because this would drive a wedge between the rate at which the Treasury can borrow and the rate of time discount.

The idea that, for some range of plausible parameter values, expansionary fiscal policy is self-financing means that for a *wider* range of parameter values, expansionary fiscal policy passes sensible benefit-cost tests. The benefits from such policy are, as before, the current-period boost to production and income from higher demand, and future-period boosts to potential output from the smaller shadow cast on future growth by a shorter and shallower downturn. The costs are the drag on future output produced by the higher taxes needed to amortize the debt incurred to finance the fiscal expansion. If fiscal expansion is self-financing, there are no costs, only benefits. And if fiscal expansion is nearly self-financing, then the increase in taxes needed to amortize the debt will be small, and so will the costs. The appendix details the arithmetic of such an extra-output benefit-cost calculation.

II. The Value of the Multiplier

Valerie Ramey (2011) surveys estimates of the fiscal multiplier and classifies them into four groups: estimates from structural models, estimates

8. How is it that a government can borrow at less than the social rate of time discount? Perhaps because government debt has unique collateralization properties that make it in some sense “money-like” (see Krishnamurthy and Vissing-Jorgensen 2012). In this case the wedge between the government borrowing rate and the social rate of time discount captures a real service flow provided to the economy by the provision of extra government debt. To the extent that the government can borrow unusually cheaply because investors are making mistakes, the welfare economics becomes complex.

from exogenous aggregate shocks (relying largely on increases in military spending associated with wars), estimates from structural vector autoregression models (VARs), and “local multiplier” estimates.⁹ She concludes (pp. 680–81) that

the range of plausible estimates for the multiplier in the case of a temporary increase in government spending that is deficit financed is probably 0.8 to 1.5. . . . If the increase is undertaken during a severe recession, the estimates are likely to be at the upper bound of this range. It should be understood, however, that there is significant uncertainty involved in these estimates. Reasonable people could argue that the multiplier is 0.5 or 2.0. . . .

Christina Romer (2011) also surveys multiplier estimates. She summarizes the evidence as suggesting a somewhat higher central tendency for estimates of the government purchases multiplier slightly above 1.5. She stresses a strong presumption that econometric estimates are likely to be lower than the constant-monetary-and-financial-conditions multiplier, which as we argue below is itself a lower bound to the current policy-relevant multiplier. As Romer (p. 11) states, concurring with Emi Nakamura and Jón Steinsson (2011): “In the situation like the one we are facing now, where monetary policy is constrained by the fact that interest rates are already close to zero, the aggregate impact of an increase in government spending may be quite a bit larger than the cross-sectional effect.”

The International Monetary Fund (IMF 2009) finds a government purchases multiplier in a broad range of post-World War II experiences similar to Romer’s central estimate. Alan Auerbach and Yurii Gorodnichenko (forthcoming) attempt to distinguish the multiplier in normal times from that which prevails when the economy suffers from slack aggregate demand. They estimate a multiplier of around 0.5 for normal times and around 2.5 when the economy is depressed.¹⁰ IMF (2010) concludes that the multiplier at the zero lower bound is more than twice what it is in normal times.

9. See, among many, many others, Ramey and Shapiro (1998), Blanchard and Perotti (2002), Gordon and Krenn (2010), Suárez Serrato and Wingender (2010), Clemens and Miran (2010), Barro and Redlick (2011), Nakamura and Steinsson (2011), Chodorow-Reich and others (2011), Romer (2011), Mendel (2012), and Ramey (2012). Moretti (2010) estimates a local multiplier that is explicitly a supply-side economic-geography concept rather than a demand-side macroeconomic concept. The relationship between economic-geography local multipliers and macroeconomic local multipliers is not clear to us.

10. See Parker (2011) on the importance of nonlinearities and on the difficulty of picking out the depressed-economy multiplier of interest here. Hall (forthcoming), however, cautions that Auerbach and Gorodnichenko’s finding “has little to do with the current thought that the multiplier is much higher when the interest rate is at its lower bound of zero . . . [for their] . . . sample surely includes only a few years when any country apart from Japan was near the lower bound.”

To summarize: the range of current multiplier estimates extends from Ramey's lowest for which "reasonable people could argue," 0.5, up to Auerbach and Gorodnichenko's estimate of 2.5, which applies when GDP is below potential such that increases in nominal spending are highly likely to show up primarily as increases in real GDP. However, it is far from clear that these estimates or the methodologies that generate them shed sufficient light on the fiscal multiplier concept relevant for our framework in section I. At present in the United States, not only is GDP below potential, but the zero lower bound constrains interest rates, and substantial frictions interfere with the functioning of credit markets. These features were seldom present during the periods and in the countries for which these multipliers were estimated.

We can use Ramey's categorization to rehearse some of the potential problems with applying these multiplier estimates from the literature to a depressed economy. First, structural model estimates are only as good as the identification of the structural model. Second, estimates based on changes in military spending will underestimate the impact of fiscal policy in a context like the present, to the extent that spending increases are associated with tax increases and Ricardian equivalence does not hold in full, or to the extent that supply constraints associated either with the rapid shift of production, heedless of efficiency, from civilian to military uses found in an emergency military mobilization, or with a high rate of resource utilization, slow output growth. Third, the identification of exogenous fiscal shocks using time-series techniques seems to us problematic: it is often difficult to identify historical events in the narrative or contemporary notes that expectations have shifted in those quarters in which time-series techniques identify "shocks" orthogonal to an information set consisting of a few lagged values.

Most promising are the estimates of "local multipliers" made by Nakamura and Steinsson (2011) and an increasing number of others. They examine differences in government spending across regions and identify a multiplier holding monetary and financial conditions constant. This literature appears to be coalescing around an estimate for such a multiplier of 1.5, although with substantial imprecision.¹¹

11. There remains some uneasiness about the interpretation of local multiplier estimates. The presence of demand spillovers across regions tends to bias such estimates down, as does the possibility that higher expected inflation rates, in the manner of Christiano, Eichenbaum, and Rebelo (2011) and Eggertsson and Krugman (2011), are a channel of transmission. Moreover, consider a permanent increase in government purchases in one region financed by taxes on all regions. Under a full Ricardian regime, such a permanent increase in spending would have no effect at all on demand and output. Yet a local multiplier study would show a considerable multiplier in both the short and the long run—an economic-geography

The principal issue in linking these estimated multipliers to the reduced-form fiscal multiplier relevant for the framework of section I is whether and to what extent the monetary policy reaction function in normal times differs from that in a depressed economy. Indeed, our suspicion is that much of the substantial variation over the past 80 years in the judgments of American economists, at least, about discretionary fiscal policy reflects changes in the nature of this function, and thus in the monetary-and-financial-conditions curve that underlies their analyses. As views of the likely slope of this function (depicted as the MP curve in figures 2 through 4 below) have changed, views of the efficacy of fiscal expansion in a depression have changed as well.

From the time of Keynes' *General Theory* to the 1960s, the default assumption was that interest rates would remain constant as fiscal policy changed, because the central bank and the fiscal authority would cooperate to support aggregate demand: fiscal expansion would be accompanied by monetary policy accommodation that produced not crowding out but crowding in. With the changes in macroeconomic thinking and the inflationary experience of the 1970s, the natural assumption in the United States came to be that the Federal Reserve was managing aggregate demand. Thus, changes in fiscal policy, just like changes in private investment demand, would be offset as the Federal Reserve pursued the appropriate balance between inflation and investment. Today, however, at least until the economy exits from the zero lower bound or cyclical unemployment drops substantially, the economy is once again in a regime in which real interest rate movements amplify rather than offset the effects of fiscal stimulus.

Consider a central bank that includes both inflation and output in its objective function, in an economy that is well modeled by the neo-Hicksian framework of Romer (2000). In such an economy, output Y and the real interest rate charged to firms r^f are jointly determined by an IS saving-investment condition and an MP monetary policy reaction function. Assume that real aggregate demand is a function of the fiscal policy impetus ΔG , the constant-monetary-and-financial-conditions multiplier μ , and r^f . An increase in government purchases in the current period from

parameter: the inverse of 1 minus the share of regional demand spent on locally produced commodities. As Mendel (2012) points out, local multiplier studies not only hold monetary and financial conditions constant; they also hold constant future fiscal conditions in the form of expectations of future broad-based taxes. To the extent that the argument against the effectiveness of expansionary fiscal policy relies on present-day reductions in spending stemming from anticipated future tax burdens, local multiplier studies will overstate the policy-relevant concept.

baseline, ΔG , would then, all else equal, raise current-period output relative to baseline according to the IS condition:

$$(8) \quad \Delta Y = -\alpha(\Delta r^f) + \mu \Delta G.$$

However, if the monetary authority responds to this expansionary fiscal policy by raising r^f or allowing it to rise, according to the following MP function,

$$(9) \quad \Delta r^f = (1/\gamma) \Delta Y,$$

then the reduced-form relationship between the fiscal expansion and the resulting difference in output from baseline is

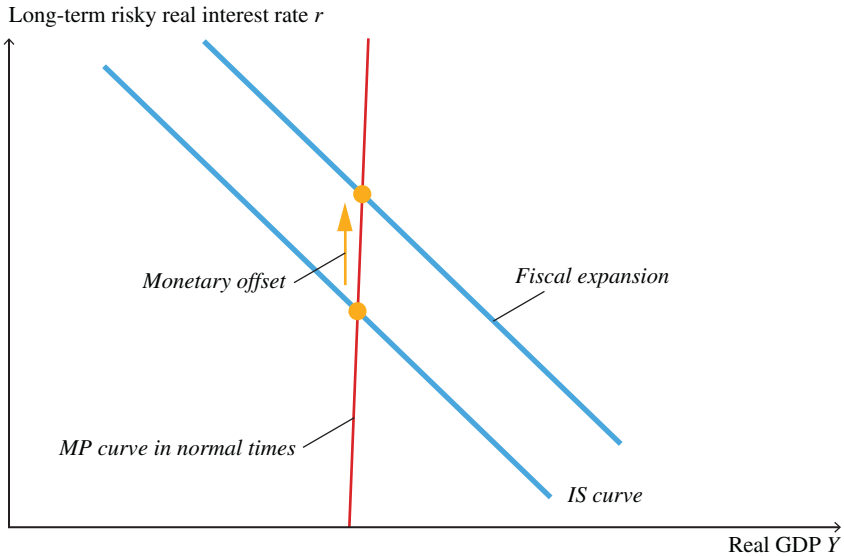
$$(10) \quad \Delta Y = \frac{\gamma}{(\gamma + \alpha)} \mu \Delta G.$$

Thus, an estimate of the multiplier over a period during which the monetary policy reaction function is characterized by a particular γ will give not the constant-monetary-and-financial-conditions multiplier μ , but rather

$$(11) \quad \mu' = \frac{\gamma}{(\gamma + \alpha)} \mu.$$

What value of γ will an optimizing central bank pick for its reaction function if, like the Federal Reserve from the end of the 1970s to the mid-2000s, it is focused on its price stability mission? The central bank will have a view about what level of Y is best suited to advance that mission over the long term. That level of Y will not be much altered by the stance of fiscal policy. The implication then is that the central bank will pick a value of γ very close to zero, and the MP curve will be nearly vertical. Whatever shocks shift the IS curve, whether fiscal policy or other factors, will then affect interest rates but will affect the level of output little if at all. Thus, in normal times the policy-relevant reduced-form multiplier μ' is likely to be small. Figure 2 illustrates this monetary offset of the fiscal expansion in normal times.

The situation is different when the economy is at the zero bound, precisely because the fiscal expansion ΔG then extends the set of economic outcomes Π attainable through monetary policy in a manner that provides access to superior outcomes previously unreachable. At the zero bound, the central bank is setting the short-term safe nominal interest rate i that it

Figure 2. IS-MP Analysis of a Fiscal Expansion in Normal Times^a

Source: Authors' model described in the text.

a. In normal times, when the economy is at or near full employment and short-term interest rates are away from their zero nominal lower bound, any attempt to increase real GDP through fiscal expansion is virtually fully offset by a rise in real interest rates.

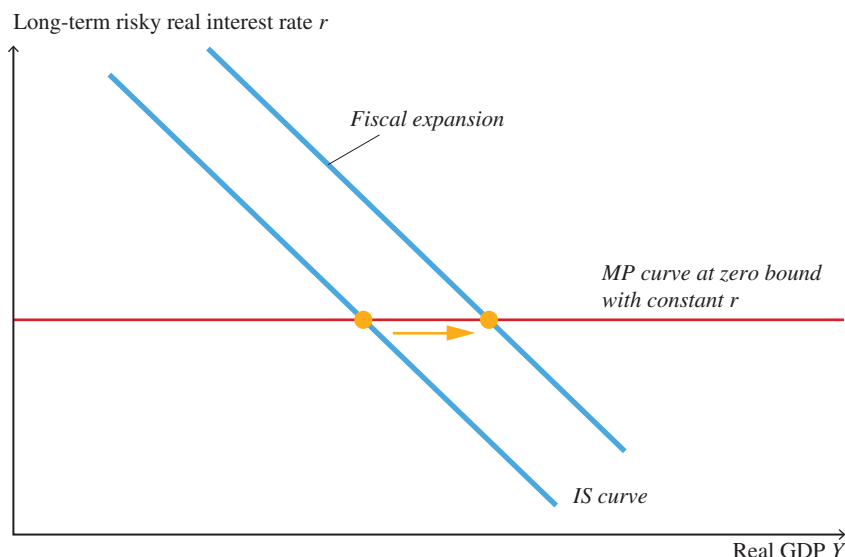
controls at zero. It would not respond to fiscal policy that boosts output by raising the short-term nominal interest rate to offset its effects, for that level of output is a previously unreachable superior outcome.

If the long-term rate to firms r^f were at a constant premium to the short-term safe nominal interest rate i , then at the zero bound the monetary policy reaction function would set a constant real rate. The MP curve would be flat, and the parameter α in equation 8 would be zero. And as in figure 3, the policy-relevant reduced-form multiplier would equal the constant-monetary-and-financial-conditions multiplier: $\mu' = \mu$.

In reality, however, there is slippage between i and r^f . The relationship between them is

$$(12) \quad r^f = i - \pi + \sigma.$$

In words, the relevant real interest rate is equal to the short-term safe rate, minus inflation, plus a spread σ —which itself has duration, risk, and default components. The inflation rate will be increasing in output: more

Figure 3. Fiscal Expansion at the Zero Lower Bound with a Constant Real Interest Rate^a

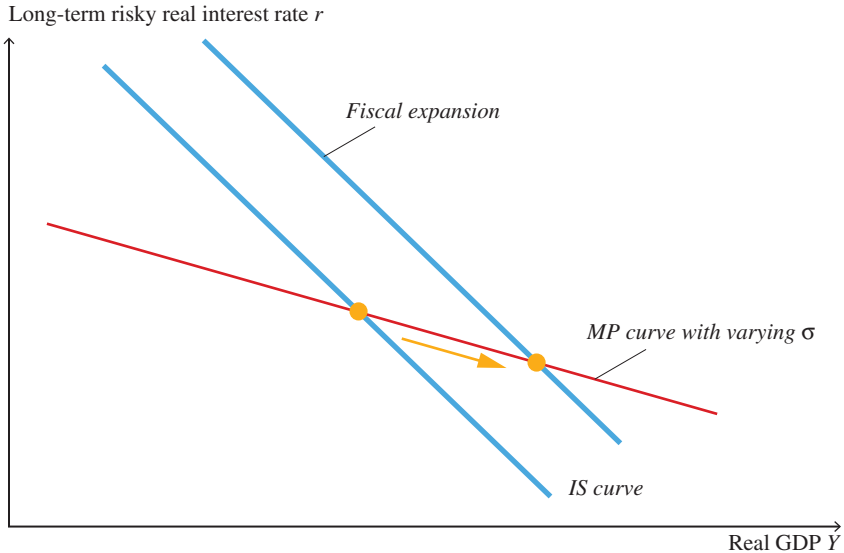
Source: Authors' model described in the text.

a. In a depressed economy where short-term interest rates are at the zero bound, a constant spread between short-term safe and long-term risky rates means the MP curve is flat. Thus, real long-term interest rates do not rise to attenuate the impact of fiscal expansion on real GDP, but neither do they fall to amplify it.

demand both raises the chances that producers will increase prices and increases how much they will raise them.¹² The interest rate spread σ , in contrast, may well be a decreasing function of output: a more prosperous economy is one with fewer defaults, and the price of bearing risk is lower because there is less risk in the economy.¹³

12. Christiano and others (2011), Eggertsson and Krugman (2012), and others point out that the impact of upward price pressure expected from expanded aggregate demand on real interest rates at the zero bound could have substantial quantitative significance. Earlier the same point had been phrased in reverse, as a fear of the potentially catastrophic consequences of deflation. See Fisher (1933).

13. The effects on duration premiums are less clear. One potential channel is that, in a depressed economy, with short-term safe nominal interest rates at their zero lower bound, if monetary authorities are willing to commit to keeping them there for a considerable period, the framework-relevant reduced-form multiplier is likely to be even larger to the extent that inflation is inertial: higher inflation in the short run due to fiscal expansion will raise expected inflation and thus lower the real interest rates expected for future periods as well. With a product-market equilibrium condition IS slope α of -0.6 as in Hall (2012), an expected duration of the zero lower bound of 3 years could double the policy-relevant reduced-form multiplier relative to the constant-monetary-and-financial-conditions multiplier.

Figure 4. Fiscal Expansion at the Zero Lower Bound with a Declining Real Interest Rate^a

Source: Authors' model described in the text.

a. If the spread σ between short-term safe and long-term risky rates is not constant but narrows as output rises, because an economy closer to full employment presents less risk to investors, then given short-term rates at the zero bound, the MP curve is not flat but downward sloping, and the increase in real GDP from a fiscal expansion is amplified.

Thus, instead of an MP curve in which increases in GDP are associated with increases in r^f , and instead of a flat MP curve, a depressed economy at the zero bound is likely to see the following relationship between interest rates and the state of the economy:

$$(13) \quad \Delta r^f = -\delta \Delta Y.$$

The multiplier estimated in that case, and the one relevant for the reduced-form framework of sections I and II, will be neither the (relatively small) normal-times reduced-form multiplier μ' nor the constant-monetary-and-financial-conditions multiplier μ , but rather

$$(14) \quad \mu^* = \frac{\mu}{(1 - \alpha\delta)},$$

and the ratio of this policy-relevant multiplier at the zero bound to the normal-times multiplier will be $\mu^*/\mu' = [1 + (\alpha/\gamma)]/(1 - \alpha\delta)$.

Figure 4 illustrates this difference between the (small) multiplier likely to be seen in normal times and the multiplier relevant at the zero bound.

Whereas the MP curve in normal times is steeply sloped upward, causing virtually all of any increase in output through fiscal expansion to be offset by a rise in r^f , the MP curve relevant for a depressed economy at the zero bound slopes downward: the stronger the economy, the lower is the real cost of capital to firms seeking to borrow.

A situation in which fiscal expansion is accompanied not by higher but rather by lower real interest rates for firms fits a scenario often mentioned by observers but rarely modeled: that of “pump priming,” a term popularized by Jacob Viner and Lauchlin Currie during the New Deal of the 1930s (Jones 1978). The claim is that private spending will flood into the marketplace and boost demand, once initial government purchases have restored the normal channels of enterprise.

Note that the presence of an exceptionally accommodative monetary reaction function at the zero bound raises the possibility that an increase in government purchases might under some circumstances be self-financing even without any hysteresis at all. At a marginal tax-and-transfer share τ of 1/3, a depressed-economy policy-relevant Keynesian multiplier μ^* of 1.5 would mean that the rise in the national debt ΔD is only half as large as the spending from an expansionary fiscal boost ΔG . A μ^* of 3 would mean that fiscal policy becomes self-financing through demand channels without resort to supply-side hysteresis. Back in 1977, Walter Heller, who had served as chairman of the Council of Economic Advisers during the Kennedy and Johnson administrations, testified before the Joint Economic Committee of Congress that reduced real interest rates brought about by monetary accommodation had raised the policy-relevant multiplier applicable to the 1964 Kennedy-Johnson tax cut enough to put it on the edge of self-financing. As Bruce Bartlett (2003, p. 5) quotes Heller:

What happened to the tax cut in 1965 is difficult to pin down, but insofar as we are able to isolate it, it did seem to have a tremendously stimulative effect, a multiplied effect on the economy. It was the major factor that led to our running a \$3 billion surplus by the middle of 1965, before escalation in Vietnam struck us. It was a \$12 billion tax cut, which would be about \$33 or \$34 billion in today's terms. And within 1 year the revenues into the Federal Treasury were already above what they had been before the tax cut. . . . Did it pay for itself in increased revenues? I think the evidence is very strong that it did. . . .

From early in the Kennedy administration through the end of 1964, the proxy for the real annual rate on 10-year Treasuries calculated by subtracting the subsequent year's inflation from the nominal rate was around 3 percent;

thereafter it dropped rapidly to around 1.5 percent. The Congressional Budget Office (CBO) was more cautious than Heller, concluding that between “25% and 75%” (Bartlett 2005, p. 5) of the static 2-year debt increase from the tax cut had been offset by the boost to output and thus to tax revenue that it had delivered.

The argument that normal-times policy-relevant fiscal multipliers should be presumed to be very small can be made more general. Optimizing central banks will be expected to offset shifts in discretionary fiscal policy—and thus lead to multiplier estimates near zero—under relatively unrestrictive conditions. Consider a government choosing monetary policy so as to achieve the best economic outcome from the set of outcomes attainable by policy Π . A change in fiscal policy from baseline would change the relationship between monetary policy and the economic outcome. But unless the change in fiscal policy opens up access to an outcome not in the set Π that is superior, or eliminates access to the best economic outcome in Π , the government will shift its monetary policy so that it still picks the same economic outcome. It will thus engage in full monetary offset.

Note that for this point to hold, the choice of monetary policy m and the choice of fiscal policy g cannot themselves be part of the outcome the government values. A central bank that values a smooth path for interest rates (as did the pre-1979 Federal Reserve) or has preferences about the size of its balance sheet (as did the Federal Reserve under Paul Volcker) will not engage in full monetary offset. Monetary and fiscal policy must enter into the central bank’s objective only through their effects on economic outcomes for full monetary offset to hold.

For these reasons it is difficult, for us at least, to consider the empirical evidence on multipliers without reaching the conclusion that the base-case multiplier of 1.0 of section I is likely to be an underestimate, and perhaps a substantial underestimate, of the policy-relevant multiplier in excess-capacity economies at the zero bound like the United States today.

III. Hysteresis

As Edmund Phelps (1972) was the first to point out, there are reasons for believing that recessions impose costs even after they end, and that a “high-pressure economy” (Arthur Okun’s term for one continuously operating at potential) has continuing benefits. It is not easy to quantify these “hysteresis effects,” in part because the factors that cause a

downturn may continue to have an impact once the downturn has ended, which is difficult to disentangle from the hysteresis effect. In this section we survey some of the evidence in an effort to come to a plausible view about our reduced-form framework parameter η , the impact of a 1-percentage-point shortfall of GDP below potential for 1 year on the subsequent path of potential output.

It would indeed be surprising if downturns did not cast a shadow over future economic activity. A host of mechanisms have been suggested, including reduced labor force attachment on the part of the long-term unemployed, scarring effects on young workers who have trouble beginning their careers, reductions in government physical and human capital investments as social insurance expenditures make prior claims on limited public financial resources, reduced investment in both in research and development and in physical capital, reduced experimentation with business models and informational spillovers, and changes in managerial attitudes.

Bottom-up evidence on hysteresis is provided by Kim Clark and Summers (1982), who documented substantial persistence in individuals' labor supply decisions and found that past work experience was a key determinant of current employment status. They concluded that this persistence of labor supply decisions meant that the hypothesis of a "natural" or non-accelerating-inflation rate of unemployment (NAIRU), as a medium-run proposition, was false. Steven Davis and Till von Wachter (2011) find that workers who lose their jobs when unemployment is high lose an extra amount, relative to when unemployment is low, equal to the present value of an extra 1.5 years of earnings in their subsequent careers—a 7.5 percent reduction in permanent earnings. At a typical average unemployment duration of 17 weeks, the aggregate demand shock associated with such a loss of employment amounts to a third of a year's earnings. This suggests a contribution to the η parameter of 0.225 ($0.075 \div 0.333$) from the labor side alone, and that only if the average duration of unemployment rapidly returns to normal levels.¹⁴

In addition to these effects on the labor side, the past several years have seen substantial shortfalls in both public and private investment.

14. Such calibration efforts are hazardous. The potential for selection effects to confound estimates is large. There is little warrant for believing that the difference between income losses following layoffs in low- and those in high-unemployment periods in the past corresponds to the effects of a shock outside the previous range like the one the U.S. economy is now experiencing.

Government nondefense capital formation in the United States is already 0.4 percentage point below its early-2008 peak as a share of potential GDP, and cuts continue. Private gross investment is still 3.5 percentage points of potential GDP below its precrisis level and has been depressed for 4 years.

The natural way to calibrate these effects on the investment side to the current downturn is to say that a 20-percentage-point-year cumulative shortfall from potential GDP has carried with it a relative decline in the capital stock equal to 14 percentage points (3.5 percentage points $\times$ 4 years) of annual potential GDP. At a marginal product of capital of 10 percent per year, that implies a 1.3 percent reduction in potential output and an investment-side contribution to η of 0.13; at a marginal product of capital of 5 percent per year, it implies a 0.65 percent reduction in potential output and an investment-side contribution to η of 0.065.

In the standard Solow growth model, the shortfall in private investment generated by the financial crisis and the recession will eventually be made up as the economy reconverges to its steady-state capital-to-output ratio. Long-term-unemployed workers who become discouraged and drop out of the labor force will reach retirement age within several decades. The long-run effects of a long, deep downturn on potential output are thus much more plausibly viewed as persistent than as truly permanent. The $1/e$ time of convergence to the steady-state capital-to-output ratio is on the order of 33 years. The average time to retirement of labor force dropouts is likely to be somewhat less. Thus, permanent-equivalent measures of the persistent effects of downturns on future potential output will be somewhat smaller. Even so, the bottom-up evidence of persistent effects of downturns on potential output indicates a value for η that is at or above the top of the range considered in section I.

Top-down evidence for hysteresis in Europe was provided by Olivier Blanchard and Summers (1986). Reacting to increases in the unemployment rate in Western Europe from the 1970s to the mid-1980s, they argued that hysteresis links between the short-run cycle and the long-run trend were key: that increases in unemployment from recessions “have a direct impact on the ‘natural’ rate of unemployment” around which an economy would oscillate. Others had argued that Western Europe’s persistently high unemployment was primarily due to rigidities in labor markets (high minimum wages, high firing costs, and the like). Laurence Ball (1997), however, suggested that the link between labor market rigidities and the transformation of cyclical into structural unemployment in Western Europe in the 1980s had been overdrawn. In his estimation, “countries with larger

decreases in inflation and longer disinflationary periods have larger rises in the NAIRU. [Measured] imperfections in the labor market [had] little direct relation to change in the NAIRU,”¹⁵ with the possible exception of an interaction between the generosity of the unemployment insurance system and the depth of the downturn.

Ball’s (1997) attribution of cross-national variation in changes in the NAIRU in the 1980s and 1990s to inadequate stabilization policy in some countries that allowed cyclical unemployment to turn structural has striking implications. He finds that in countries that pursue long, slow rather than short, sharp disinflations—with an active pursuit of disinflation on the order of 4 years—effectively all of the cyclical decline in employment becomes a permanent decline. Four percentage-point-years of a negative shock thus produces a 1 percent fall in potential output, for an η of 0.25.

Findings similar to those of Ball (1997) are reported in IMF (2009), which examines the effects of demand shocks produced by financial crises at a 7-year horizon. In that study of the aftermath of 88 financial crises in the past two generations, each output decline of 1 percent of GDP in the short-run response to a financial crisis is associated on average with a 1 percent shortfall of GDP from its precrisis trend. If the “short run” during which output is depressed because of inadequate demand is 3 years, this result is consistent with an η of 0.33.¹⁶

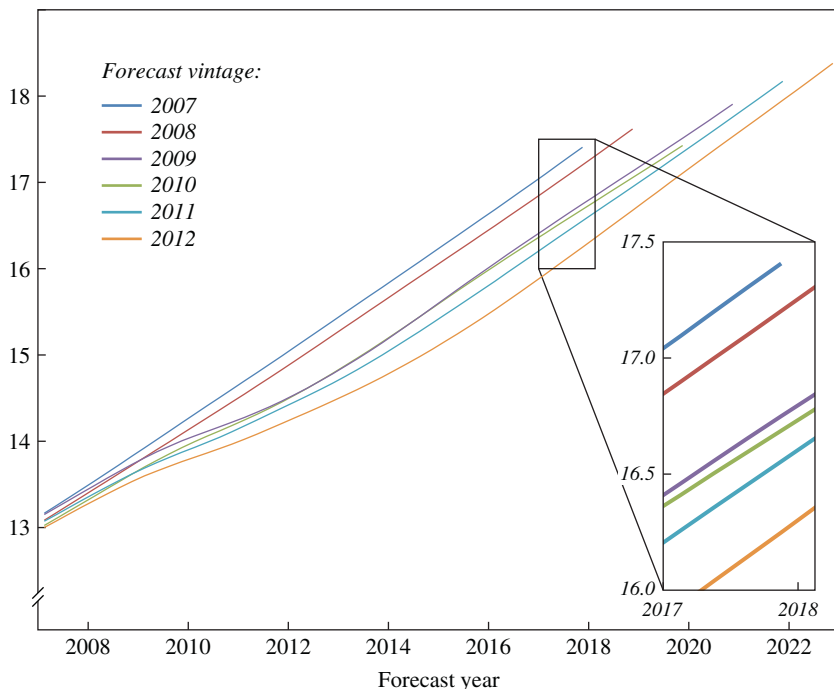
A second form of top-down evidence is provided by professional economic forecasters. As a group, they do not appear to hold to the position that the current economic downturn will have no or small effects on the growth path of U.S. potential output. Instead, their recent revisions of their projections for the next decade implicitly incorporate very substantial hysteresis effects. To take one prominent example: between January 2007 and January 2009, as the economy slid into its deep, financial crisis-driven recession, the CBO marked down its estimate of potential GDP for the end of 2017 by 4.2 percent (figure 5). The CBO took some heart from the end of the recession in late 2009, and in its January 2010 forecast revision

15. Ball (1997, p. 168). See, in addition, Stockhammer and Sturn (2012), who also conclude that the degree of labor-side hysteresis is likely to have only weak connections with labor market institutions but a rather strong association with the persistence of high unemployment and the failure of activist stabilization policies to quickly fill the output gaps created by downturns. In their results, hysteresis has “strong [associations with] monetary policy, and . . . [perhaps] the change in the terms of trade, but weak (if any) effects of labour market institutions during recession periods. Those countries which more aggressively reduced their real interest rates in the vulnerable period of a recession experienced a much smaller increase in the NAIRU. . . .”

16. Also consistent is Romer (1989), who argues that the output effects of demand shocks are very long lasting.

Figure 5. Recent CBO Forecasts of Potential GDP

Trillions of 2005 chained dollars



Source: Congressional Budget Office.

it raised its estimate of end-of-2017 potential GDP by 0.4 percent. Then, over the next 2 years to January 2012, the CBO—in near lockstep with private forecasters—lowered its forecast of end-of-2017 potential GDP by an additional 3 percent. Thus, as of the beginning of 2012, the CBO had marked down its estimate of potential GDP 5 years hence by a cumulative 6.8 percentage points. Were that markdown to be interpreted as the result simply of the 20-percentage-point-year output gap to the present, it would correspond to an η of 0.34. Even if that markdown were based on a belief that the economy has so far experienced only half of the cumulative gap relative to potential output that will ultimately result from this episode, that would correspond to an η of 0.17.

It is possible that these revisions reflect not a belief in hysteresis but merely the recognition that previous forecasts of potential output were too high. However, an elementary signal extraction point rebuts this interpretation. When observing a noisy series that has a permanent

component, an observation lower than the current estimate of the permanent component leads a rational forecaster to reduce his or her estimate of that permanent component. However, one should not reduce one's estimate of potential output if lower-than-previously-expected levels of production are associated with lower-than-previously-expected levels of inflation. Estimates of potential output are conceptually based not on quantities alone, but on quantities and prices. Typically, the bad news that leads to a marking down of potential output is not news that output is lower than, but rather news that output and inflation together are above, their anticipated co-movement line. Such news is not in evidence.

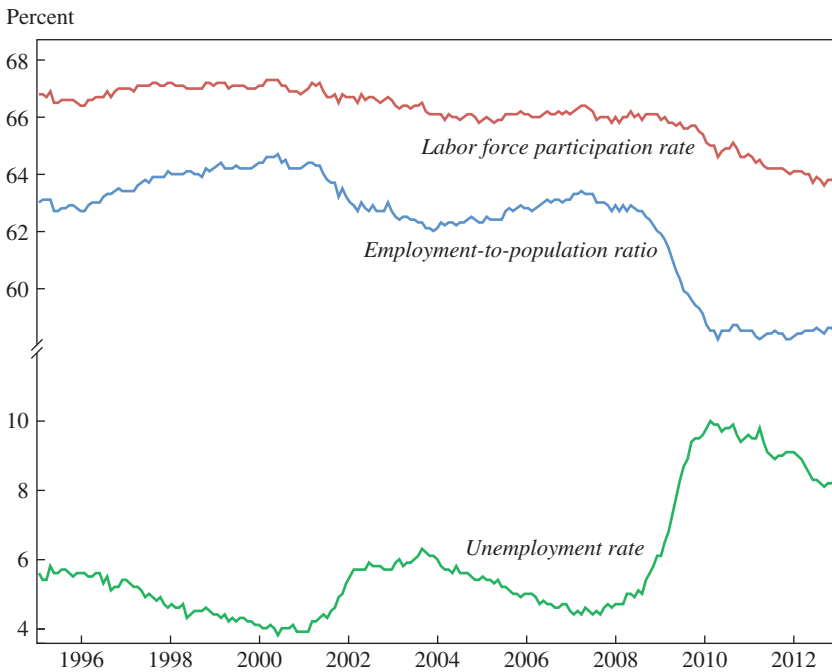
Blanchard and Summers's (1986) line of thought was that significant hysteresis was a uniquely European phenomenon. Their model carried the implication that the United States was likely to be largely immune from permanent labor-side effects of what was originally transitory cyclical unemployment.¹⁷ They stressed the "insider-outsider" wage-bargaining theory of hysteresis: workers who lose their jobs no longer vote in union elections, and so union leaders no longer take their interests into account in negotiations, focusing instead on higher wages and better working conditions for those still employed. Since union strength and legal obligations on employers to bargain were much weaker in the United States than in Europe, insider-outsider dynamics generated by formal labor market institutions seemed to give the United States little to fear.

However, the labor market dynamics of the past two and a half years raise the possibility that the United States is not so immune after all from the considerations raised by Blanchard and Summers (1986). Rather, a transformation of cyclical into structural unemployment may be under way in the United States today, as the pace of real GDP growth during the current recovery is no greater than the precrisis trend growth rate of potential output, so that the output gap remains large.

Here it is worth noting the divergence between the behavior of the measured U.S. unemployment rate and the behavior of the measured U.S. adult employment-to-population ratio over the past two and a half years. From the late-2009 peak in the unemployment rate until April 2012, the civilian employment-to-population ratio fell by only 0.1 percentage point, the civilian adult labor force participation rate by a more substantial 1.4 percentage points, and the unemployment rate by an even larger 1.9 percentage points, from 10.0 percent to 8.1 percent (figure 6).

17. An alternative also put forward by Blanchard and Summers (1986) focuses on how the long-term unemployed become detached from the labor market. See Granovetter (1973) and especially Layard, Nickell, and Jackman (2005).

Figure 6. Labor Force Participation Rate, Employment-to-Population Ratio, and Unemployment Rate, 1995–2012



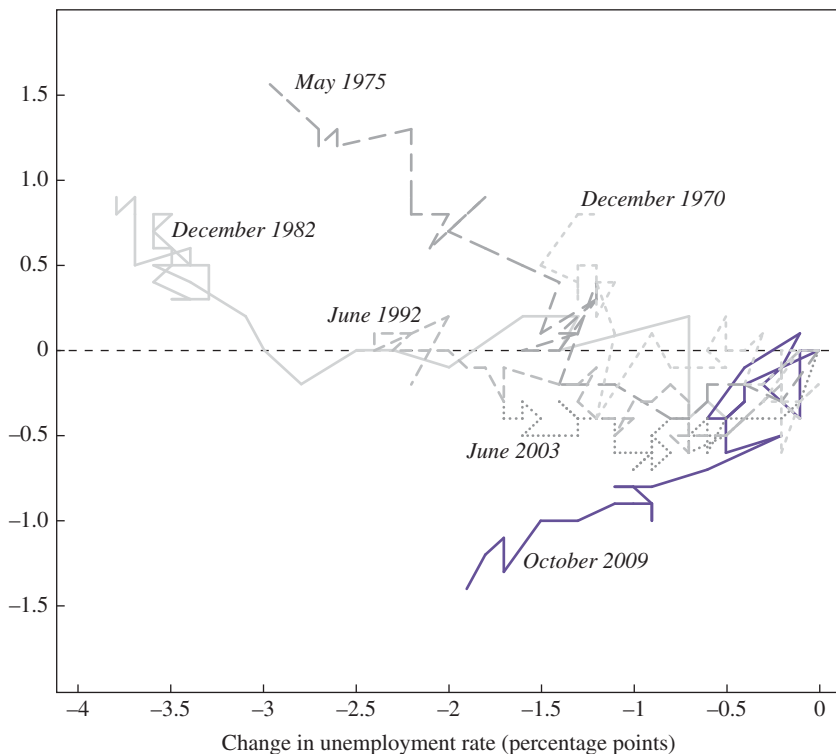
Source: Current Population Survey.

Such a divergence between the unemployment rate and the employment-to-population ratio is unprecedented in the United States. The years immediately following the 1970, 1975, and 1982 unemployment rate peaks saw strong recovery in the labor force participation rate, and the 1992 and 2003 unemployment rate peaks were followed by effectively flat labor force participation rates and very slow eventual recoveries. Only after the 2009 unemployment rate peak has the civilian labor force participation rate continued to decline, and indeed to decline by enough to offset the effects of the falling unemployment rate, leaving the employment-to-population ratio virtually unchanged from the low point reached at the end of the recession (figure 7).

Since the late 1990s, the retirement of many members of the baby-boom generation has led to lower employment-to-population ratios for a given unemployment rate. But this is a slow-moving generational trend, amounting to a fall in labor force participation on the order of 0.05 percentage

Figure 7. Changes in Unemployment and Labor Force Participation Rates after Cyclical Peaks^a

Change in labor force participation rate (percentage points)



Source: Authors' calculations from Current Population Survey data.

a. Each line plots the month-to-month changes in the two measures for a single recession, beginning with the month (labeled) in which the unemployment rate peaked.

point per year. The total reduction in labor force participation since the end of the recession is thus an order of magnitude too large to be attributed to this phenomenon alone.¹⁸ Moreover, there are counteracting pressures stemming from the financial crisis that should tend to raise labor force participation: one would expect many middle-aged Americans whose wealth (housing or financial, or both) has been reduced by the crisis to delay retirement. Indeed, there are signs of such a wealth effect at work in the increasing employment of those past retirement age since 2007.

18. See Daly, Hobijn, and Valetta (2011). There is a potential argument for an interaction effect, however: perhaps the older labor force of today is more likely to be induced into early retirement by the experience of unemployment.

Consider a counterfactual in which the unemployment rate had followed its actual trend but the labor force participation rate had remained at its same level between October 2009 and April 2012, rather than falling by 1.4 percentage points in those 30 months as it did. The supply of workers in America today is 2.2 percent lower than in that counterfactual baseline. Under the assumption that potential output scales one for one with the labor force, such a reduction in labor supply implies a 2.2 percent reduction in potential output. Assuming instead a potential-output production function with a labor share of 0.65, the reduction in potential output would be 1.4 percent.

From the start of 2008 through the end of 2011, the cumulative shortfall of real GDP from the Congressional Budget Office's potential GDP series amounted to 20.5-percentage-point-years. Under the assumption that potential output scales one for one with the labor force, dividing 2.2 percent by 20.5-percentage-point-years yields an η of 0.107; assuming instead that potential output scales with a labor share of 0.65 gives an η of 0.07. Moreover, this calculation assumes that the NAIRU has remained unchanged over the past 5 years. Christina Romer (2012) documents, however, that the NAIRU estimates of the CBO, the Federal Open Market Committee, and the Survey of Professional Forecasters have been raised since 2007 by 0.8, 0.7, and 1.2 percentage points, respectively.¹⁹ A counterfactual in which the NAIRU had remained at its 2007 rate would produce a potential labor force at full employment 3.0 percent larger than the current situation, which would imply correspondingly higher values of η .

The U.S. economy in the aftermath of the 2008–09 crisis thus appears not to be repeating the exceptional rapid rebound that used to distinguish it from the sclerotic Western Europe analyzed by Blanchard and Summers (1986). Instead it seems to be following much more closely the typical post-financial crisis pattern found by IMF (2009). In their sample, 7 years after the crisis, real GDP on average was some 10 percent below its precrisis trend.²⁰ Both the capital stock and employment were substantially depressed below their precrisis trends, with shortfalls relative to previous trends in total factor productivity as well. In particular, IMF (2009, pp. 4–5) found:

—There was, on average, no recovery to trend from the level relative to trend of the short-run output decline: “the path of output tends to

19. The CBO's estimates are found in its *Budget and Economic Outlook*, various issues; those of the Federal Open Market Committee in its Summary of Economic Projections, various issues; and those of the Survey of Professional Forecasters in Federal Reserve Bank of Philadelphia (2011).

20. The IMF is relatively strident on this point. It writes of “sobering implications” of the analysis and praises “forceful macroeconomic policy response[s] . . . in the form of substantial fiscal and monetary stimulus.”

be depressed substantially and persistently . . . with no rebound on average to the precrisis trend.”

—Crises that did not generate large output declines in the short run tended not to generate large shortfalls relative to trend at the 7-year horizon: “what happens to short-run output is also a good predictor of the medium-term outcome.”

—The economies that did approach their precrisis trend growth path in recovery tended to be those that had applied substantial macroeconomic stimulus immediately after the crisis: “although post-crisis output dynamics are hard to predict, the evidence suggests that economies that apply counter-cyclical fiscal and monetary stimulus in the short run after the crisis tend to have smaller output losses” relative to trend at the 7-year horizon.

The historical evidence on the existence of hysteresis is thinner than one would wish, as is inevitable when one is attempting to generalize from a few previous episodes. Thus, any conclusions must be weak and tentative. The question of how large a shadow is cast on future potential output by a deep cyclical downturn rests on a few historical cases: the experience of the United States and Western Europe in the Great Depression, the long Western European downturn of the late 1970s and the 1980s (comparing both Europe with the United States and the European countries with each other), and Japan’s “lost decades” starting in the 1990s. In the United States, moreover, the Great Depression was followed by the great boom of total mobilization for World War II, so that if the Great Depression did cast a shadow, it was erased by the war.

Perhaps the recent departure of the unemployment rate and the labor force participation rate from their earlier historical pattern of co-movement will turn out to be a transitory cyclical anomaly. Perhaps in the next few years the economy will quickly rebound to its pre-2008 path of potential output growth. But our reading of the remaining cases—the experience of Western Europe since the late 1970s and Japan during the 1990s and after—provide strong reason to presume that hysteresis effects on the order of those in table 2 are more likely than not to be a reality. In that case the standard call for further research in this area becomes urgent.

IV. Conclusion

Real interest rates on Treasury securities have fluctuated within a relatively narrow range throughout their history, except for the few years of the Volcker disinflation of the early 1980s. Rates in this historical range, in

a depressed economy at the zero lower bound, with even a modest short-run government purchases multiplier μ and a small hysteresis parameter η , generate as a matter of arithmetic the conclusion that expansionary fiscal policy does not impose a future fiscal burden. Moreover, as the appendix shows, even when expansionary fiscal policy fails to be self-financing in these circumstances, it is still likely to pass a sensible extra-output benefit-cost test, at least as long as there is no substantial wedge between the government's real borrowing cost and the real social rate of time discount.

Sections II and III made the case that the short-run reduced-form policy-relevant fiscal multiplier μ is likely to be substantial enough *in a depressed economy*, and that hysteresis effects η are likely to be present. And there is today no sign of a large wedge between the government's real borrowing cost and the real social rate of time discount.

It is important to stress that our argument does not justify unsustainable fiscal policies, nor does it justify delaying the passage of legislation to make unsustainable fiscal policies sustainable. If committed spending and committed revenue plans are inconsistent, adjustments will be necessary. Nothing in our analysis calls into question the widely held proposition that it is desirable for those adjustments to be committed to sooner rather than later. Indeed, the sooner that is done, the less likely is the emergence of the wedge between government borrowing costs and the social discount rate that would make expansionary fiscal policy unwise even in a depressed economy. Expansionary fiscal policy is more likely to be self-financing when there is confidence in long-run fiscal balance than otherwise.

Three crucial questions confront any attempt to draw policy implications:

—Doesn't the argument prove too much? Can it be the case that most governments at most times can take on increased debt, relying on the benefits of induced growth to pay it back?

—Is the kind of temporary fiscal stimulus envisioned in our model feasible in the world, or does it inevitably, in reality or perception, become at least quasi-permanent, thus amplifying debt-servicing costs without amplifying the output benefits?

—Third, whatever the merits of fiscal stimulus, should not monetary policy be relied on as an alternative and superior instrument?

We briefly consider each of these questions in turn.

On the first question, it surely cannot be the case that more expansion is desirable most of the time. We have stressed our belief that, outside of extraordinary downturns where the zero lower bound constrains interest rates, the right assumption is that the fiscal multiplier is likely to be small.

Increases in demand run up against supply constraints,²¹ even when they are not offset by monetary policy. And in the normal-times case of a small policy-relevant multiplier, judgments about fiscal policies should be made on allocative rather than stabilization policy grounds. As a corollary, even in depressed economies, expansionary fiscal policy surely should not be pursued without limit.

With regard to the second question, the premise of our analysis is that expansionary fiscal policy can be both timely and temporary. Thus, it makes a case only for as much fiscal stimulus as can be delivered in a timely and temporary way. If, because of political frictions, stimulus will not in fact be temporary, or if there are substantial lags in its implementation, the calculus of costs and benefits is altered. Is temporary stimulus inconsistent with belief in long-run consolidation? It is possible that short-run fiscal expansion undercuts the credibility of long-run fiscal consolidation. It is also possible that, in a world with limited political energy and substantial procedural blockages, any effort toward one objective compromises the other.

Our reading of the recent U.S. experience is encouraging as to the feasibility of significant timely and temporary stimulus—contrary to Taylor (2011), Juan Carlos Suárez Serrato and Philippe Wingender (2010), and others who suggest that a substantial fraction of the fiscal stimulus enacted in the 2009 recovery act translated rapidly into increased spending and was not offset by triggered changes in state and local fiscal policy. There is also experience with phased-in long-run deficit reductions (for example, the 1983 bipartisan agreement on the Social Security recommendations of the Greenspan Commission). The recent U.S. experience also suggests that fiscal stimulus can be reversed: certainly whatever stimulus was provided by the 2009 act already has been.

But even if it is granted that stimulus can be timely and temporary, the question of how large it can be while preserving these attributes remains for future research.²² And as Carlo Cottarelli (2012) warns, countries that

21. Note that Gordon and Krenn (2010) find a multiplier of 1.88 for the pre–Pearl Harbor mobilization for World War II at the zero nominal bound when they end their sample in the still demand-constrained first half of 1941, but of only 0.88 when they end their sample at the end of 1941, when supply constraints begin to bite. This feature does not make it into modern models. As Hall (forthcoming) comments, “The simple idea that output and employment are constrained at full employment is not reflected in any modern model that I know of. The cutting edge of general-equilibrium modeling—seen primarily in the DSGE models popular at central banks around the world—incorporates price and wage stickiness that makes supply quite elastic both above and below full employment.”

22. See Erceg and Linde (2010) on the nonlinearity of responses to fiscal expansion at the zero bound.

commit to short-term deficit reduction as a down payment on a move to long-term sustainability may find that

growth slows more than expected . . . [they are] inclined to preserve their short-term plans through additional tightening, even if it hurts growth more . . . my bottom line: unless you have to, you shouldn't . . . interest rates could actually rise [even] as the deficit falls . . . [if] growth falls enough as a result of a fiscal tightening.

On the third question, our analysis has taken it as given that at the zero bound, monetary policy does not change when fiscal policy is altered. Central banks, however, do have room for maneuver, both in their ability to operate directly on a wider range of financial instruments than they use in normal times, and in their ability to precommit policy. As a matter of logic, it is possible that increased fiscal actions will call forth a contractionary monetary policy response by causing central banks to use these tools less expansively. Perhaps, then, as Gregory Mankiw and Matthew Weinzierl (2011) assert, arguments for fiscal expansion in a depressed economy are even better arguments for monetary expansion.

On the other hand, in the United States the Federal Reserve has sought to encourage short-run fiscal expansion. There appear to be limits to the efficacy of nonstandard monetary measures and to the willingness of central banks to expand their balance sheets in order to engage in them. And expansionary fiscal policies may well both support and call forth a more expansionary monetary policy response by, for example, raising the credibility of commitments to monetary expansion after the economy has recovered, or increasing the extent of debt monetization.

It seems to us that, especially if fiscal policy is self-financing, it will be appropriate to include it in the instrument mix, for several reasons. First, given model and parameter uncertainty, diversification among policy instruments is appropriate, as William Brainard (1967) suggested long ago. Second, nonstandard monetary policies at the zero bound are perceived by central banks as carrying substantial costs or risks if engaged in on a large scale—hence central banks' hesitancy at undertaking them. Third, expansionary monetary policies carry costs not represented in standard models, including distortions in the composition of investment, impacts on the health of the financial sector, and impacts on the distribution of income. And fourth, history suggests a tendency for low-interest-rate environments to give rise to asset market bubbles, which economists and policymakers today fear more than they did even half a decade ago. Together these considerations indicate that monetary policy cannot bear all the burden. There is thus a strong case for expansionary fiscal policy in a depressed economy.

APPENDIX

An Extra-Output Benefit-Cost Test

If expression 7 in the text does not hold and the government borrowing rate exceeds or will exceed the critical value, then determining the desirability of expansionary fiscal policy calls for a benefit-cost calculation. It is appropriate to weigh present benefits from expansionary fiscal policy against future costs. A natural quantity to examine for such a benefit-cost calculation is the present value of the change in future output: the summed, discounted effects on present and future GDP of contemporary transitory fiscal expansion.²³

Call these effects ΔV . Then, in terms of the framework of section I, where ΔY_n is the impact of the transitory fiscal expansion ΔG on present-period output and ΔY_f is the impact on potential output in a representative future period,

$$(A.1) \quad \Delta V = \Delta Y_n + \frac{\Delta Y_f}{r - g},^{24}$$

where r is in this case the real social rate of time discount, which we identify here with the real government borrowing rate.

Assume that the appropriate long-run measure of r is or will rapidly normalize to a value larger than the growth rate of the tax base g . The economy is thus dynamically efficient. If the economy is not dynamically efficient, then there is no benefit-cost calculation to perform: expansionary fiscal policy is worthwhile.

Fiscal expansion has benefits in terms of higher GDP in the short run through the multiplier. It has benefits in terms of higher future potential output in the long run through the avoidance of hysteresis. These benefits are counterbalanced by the supply-side drag on future potential output from higher tax rates needed to raise the revenue to amortize the higher debt burden.

Equation A.1 assumes that the long-term effects of fiscal expansion, both through avoiding hysteresis and through debt amortization, are truly

23. The change in the present value of output can, of course, be questioned as a welfare measure. In contexts like the present, however, we suspect that the social value of the leisure of the currently unemployed is low, and that society attaches a high value to the extra output gained in the future by, for example, avoiding cutbacks to innovation spending or by avoiding labor force withdrawal by those who after a long spell of unemployment retire or apply for disability. See Krueger and Mueller (2011), Gordon (1973), Granovetter (1973), and Gordon (2011).

24. In this equation and throughout the appendix we suppress a “length-of-short-run” parameter in order to make the notation less cumbersome.

permanent and scale with economic growth. Thus, ΔV is calculated by discounting ΔY_f at the rate $r - g$. If the effects are long-lasting but not truly permanent, the appropriate discount factor in the analogue of equation A.1 would be higher, but the basic logic of the argument would remain the same: there are short-term benefits and both short- and long-term costs, with the long-term costs attenuated to the extent that the wedge between the borrowing costs and the growth rate of the tax base is relatively low.

The impact ΔY_n of the transitory contemporary fiscal expansion ΔG on current-period output is as given by equation 1 in the text. The full impact ΔY_f on potential output in a representative future period is more complex. It has two components. The first is the positive impact $\eta \Delta Y_p = \eta \mu \Delta G$ from the lessened shadow cast by the downturn on future potential output. The second is the burden imposed on future GDP by the cost of amortizing the debt incurred to finance the fiscal expansion. This second supply-side cost component depends on two factors: (i) the additional debt ΔD that must be amortized, multiplied by (ii) the disincentive effect on potential output from the higher future taxes needed to fund each dollar of amortization; we model this second factor with the parameter ξ , which represents the reduction in future potential output from raising an additional dollar of revenue. However, these costs are themselves partially offset by another supply-side effect: by avoiding or reducing hysteresis, higher current-period GDP allows the burden of amortizing the preexisting costs of government to be spread over a larger tax base, and so allows for lower tax rates and thus further raises future potential output.

If raising an additional dollar of net tax revenue in the representative future period has disincentive effects that reduce future-period GDP by ξ , then the effect on future-period real GDP is

$$(A.2) \quad \Delta Y_f = \{ \eta \mu - \xi [(r - g)(1 - \mu \tau) - \tau \eta \mu] \} \Delta G.$$

We assume the normal-case value of ξ to be 0.25 and the extreme-case value to be 0.5.

Discounting equation A.2 back to the present and adding it to equation 1 then produces the net effect of contemporary transitory expansionary fiscal policy on the present value of real GDP:

$$(A.3) \quad \Delta V = \left\{ \mu + \frac{\eta \mu}{r - g} + \frac{\xi}{r - g} [\eta \mu \tau - (r - g)(1 - \mu \tau)] \right\} \Delta G.$$

The first term within the braces on the right-hand side of equation A.3, μ , is the multiplier term. The second, $\eta \mu / (r - g)$, is the hysteresis term: the

smaller long-term shadow cast by a smaller downturn. The third term is the impact on future potential output of the net burden of additional debt. It is equal to the net impact on government cash flow, from the left-hand side of equation 6, multiplied by ξ , which captures the supply-side benefits to output from lower tax rates, expressed as a present value through division by $r - g$. This third term is composed of two subterms: $\xi\tau\eta\mu/(r - g)$ and $-\xi(1 - \mu\tau)$. The first subterm is the Blanchard and Summers (1987) term: the effect on potential output from lower tax rates made possible by the counterhysteresis effects of the fiscal expansion ΔG on potential output. The second subterm is the burden of amortizing the extra debt needed to finance the fiscal expansion ΔG . Even if this third term is negative and fiscal policy is not self-financing, expansion still passes the extra-output benefit-cost test if the first two terms are large enough to more than counterbalance it.

We draw five significant lessons from equation A.3:

- A fiscal expansion's effects are as much long-run as short-run.
- In a nondepressed economy, fiscal policy is highly likely to fail its benefit-cost test (equation A.3) because the multiplier μ is likely to be near zero.
- Even in the absence of hysteresis, fiscal policy may pass its benefit-cost test.
- Failure of the benefit-cost test in a depressed economy seems to require a high disincentive coefficient ξ .
- If interest rates substantially exceed the social rate of time discount, fiscal policy will fail its benefit-cost test.

The first lesson follows from observing that in equation A.3 only the initial term μ is a short-run term. Even outside of the consequences for cash flows, long-run benefits are a factor $\eta/(r - g)$ greater than short-term benefits. For the central case of table 2, with $\eta = 0.05$ and $\mu = 1.0$, this ratio of short- to long-term benefits is 1.7 at the critical real interest rate of $r = 5.77$ percent per year. Expansionary fiscal policy thus should not be analyzed as if pursuing it removes political-economic focus from the long run.

As with all present-value calculations at interest rates not too much larger than growth rates, a large proportion of the value comes from the distant future. If we impose the condition that our forecasting horizon ends 25 years into the future, on the grounds that the world more than a generation hence is likely to be different from the world of today in an “unknown unknowns” fashion, the ratio of long-run to short-run benefits falls to 1.14. But it is not just the long-run benefits of current expansionary policy from the counterhysteresis effect that are subject to exhaustion when a truly new deal is dealt; a truly new deal might well alter government financing burdens as well.

Our second lesson is that in a nondepressed economy, the policy-relevant reduced-form multiplier is likely to be small, and thus fiscal policy is highly likely to fail the benefit-cost test. The positive terms in equation A.3 are all linear in μ and thus shrink with μ . But the negative term $\xi(1 - \mu\tau)$ is not linear in μ and does not become small. The multiplier μ relevant for equation A.3 is a reduced-form multiplier inclusive of monetary offset. It is not the multiplier holding real or nominal interest rates constant. It is not even the multiplier holding the monetary base or the money stock constant. It is the multiplier taking into account whatever the typical monetary policy reaction function to macroeconomic news is.

In normal times that inclusive-of-monetary-offset multiplier is small. The central bank will almost invariably have strong views about what course of real aggregate demand is appropriate given its long-run price stability objectives. The central bank will be uninterested in having real demand pushed off what it regards as the appropriate path by the actions of any other agencies of government. It will thus attempt to offset whatever effects expansionary fiscal policy has on aggregate demand. And because central banks can work inside the discretionary fiscal policy decision loop of legislatures and executives, they will do so.

In a depressed economy, things are different. With interest rates at the zero bound, the central bank may lack the power to manage aggregate demand by itself without pushing nonstandard monetary policy beyond the limits it regards as plausible. And even if the central bank believes that it has the power, it may lack the will—and may well lack the formal legal authority—to undertake nonstandard policy measures that might be better classified as quasi-fiscal policies.

If, in a depressed economy, a central bank possesses both the power and the will to target real aggregate demand and offset any effects of fiscal expansion, then the policy-relevant multiplier μ in equation A.3 will be sufficiently small that expansionary fiscal policy fails to pass its benefit-cost test. But if the central bank lacks either the power or the will to do so, our argument applies. The fact that expansionary discretionary fiscal policy fails the benefit-cost test of equation A.3 in normal times carries no implications for the test in a depressed economy.

Our third lesson is that even in the absence of hysteresis effects, discretionary expansionary fiscal policy may well pass its benefit-cost test. In the absence of hysteresis effects, when $\eta = 0$, equation A.3 becomes

$$(A.4) \quad \Delta V = [\mu - \xi(1 - \mu\tau)]\Delta G.$$

This expression is positive when

$$(A.5) \quad \mu > \frac{\xi}{1 + \xi\tau}.$$

For a tax-and-transfer share τ of 1/3, a multiplier μ of 0.5 produces a positive extra-output benefit-cost test for any ξ less than 0.6:

—A μ of 1.5 produces a positive benefit-cost test for any ξ less than 3: a ξ of 3 would mean that the economy is so far to the right on the Laffer curve that the marginal dollar raised from taxes reduces potential output by \$3.

—A μ of 1 produces a positive benefit-cost test for any ξ less than 1.5.

—Even a μ of 0.5 would require a ξ of 0.6, which seems unlikely: other North Atlantic countries have significantly higher values of τ with no clearly visible signs of such severe effects of taxes on potential output.

Our fourth lesson is that adding in hysteresis effects through a positive value of η makes the arithmetic of the benefit-cost test of equation A.3 even more compelling. The analogue of expression A.5 then becomes:

$$(A.6) \quad \mu > \frac{\xi}{[1 + \xi\tau + \eta(1 + \xi\tau)/(r - g)]}.$$

For temporary expansionary fiscal policy to fail its benefit-cost test with even very moderate multiplier and hysteresis effects, the requirements are stringent. For τ of 1/3, g of 2.5 percent per year, μ of 0.5, η of 0.05, and r of 6 percent per year, temporary fiscal expansion fails its benefit-cost test only if ξ is greater than 10.

This leads to the fifth and last lesson: Only a small value of μ is typically needed in expression A.6 for expansionary fiscal policy to pass the benefit-cost test, because the critical value of μ is reduced by the hysteresis term in the denominator, and because the presence of $r - g$ can make this term large. Any set of parameter values in which $\eta/(r - g)$ is nonnegligible makes the critical value of μ small. Thus, the benefit-cost test is likely to be passed unless $r - g$ is relatively large—and in this case r is not the real social rate of time discount but instead the real Treasury borrowing rate. It follows that discretionary fiscal policy in a depressed economy is most likely to fail its benefit-cost test if there is a wedge between the real Treasury borrowing rate (which determines the burden of the debt) and the social rate of time discount (which determines the multiple at which future benefits and costs are capitalized). For a wedge ρ between the real social rate of time discount r and the government's real borrowing cost $r + \rho$, the benefit-cost calculation in equation A.3 becomes

$$(A.7) \quad \Delta V = \left[\mu + \frac{\eta\mu}{r-g} + \frac{\xi\eta\mu\tau}{r-g} - \frac{(r+\rho-g)(1-\mu\tau)}{r-g} \right] \Delta G.$$

The costs in the final term on the right-hand side are then amplified by the factor $(r + \rho - g)/(r - g)$, while the benefits in the first three terms stay the same as they were in equation A.3. A government that must borrow at the terms of a present-day Greece or Spain—or that fears that even marginal additional borrowing will produce a market reaction that will force it to borrow on such terms—will find the arithmetic of expansionary fiscal policy unpleasant indeed. But there is no such wedge for the United States today. Nor are there any visible signs in asset values that the future emergence of such a wedge is priced into today's markets, at any detectable probability.

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Comments and Discussion

COMMENT BY

MARTIN FELDSTEIN Bradford DeLong and Lawrence Summers have written a paper that usefully advances the analysis of fiscal deficits, especially in the unusual conditions of the type that have recently prevailed. Their paper takes the analysis of fiscal stimulus programs beyond the commonly asked question, “Does it work in raising GDP?” to the more appropriate analysis of whether the benefits of a fiscal stimulus outweigh the economic costs.

The authors identify three components of that cost-benefit analysis: the basic multiplier measuring the change in short-run GDP due to the increased fiscal deficit, the deadweight cost of raising funds to finance the resulting future debt, and the potential counterhysteresis effect of the increased GDP on human capital and business investment. They are careful to emphasize that their results apply only to the very unusual condition of a deep recession with the Federal Reserve’s policy interest rate constrained by the zero lower bound. I will begin by commenting on the cost of raising funds to finance the increased debt and will then turn to the measurement of the multiplier and the hysteresis effect.

THE COST OF INCREASED NATIONAL DEBT The paper lays out the possibility of what others have called a “fiscal free lunch,” in which the economic cost of raising funds is zero or negative, appearing to make any kind of government deficit creation worthwhile no matter how small the multiplier may be. If only that were true! There are so many good things that could be done if the economic cost of increasing the national debt were actually zero or negative. Extending the tax cuts that are due to expire at the end of 2012 would have favorable incentive effects in addition to any Keynesian demand effects. Reducing the mortgage debt of homeowners who owe more than their homes are worth not only would

help those individuals but also would remove the housing debt overhang as a barrier to a robust recovery. There are so many appealing things to do when government money is essentially free.

It is important to understand what causes the fiscal free lunch in the authors' analysis and to recognize several other considerations that they have omitted that increase that cost of funds. DeLong and Summers conclude that government financing today, with the real interest rate on long-term Treasury bonds below the real growth rate of the economy, is costless because, under that condition, the denominator of the debt-to-GDP ratio rises more rapidly than the numerator. Specifically, the real annual interest rate on 30-year Treasury inflation-protected securities as of early 2012 was just 0.80 percent, below the 2.5 percent long-term real GDP growth rate projected by the Congressional Budget Office. The debt ratio therefore declines without the need to raise future taxes.

But DeLong and Summers's focus on interest payments ignores four important costs of increasing the national debt even under the special economic circumstances that their paper assumes. When these are taken into account, the cost of increasing the national debt is potentially much greater than just the real net interest payments.

The first cost is that due to foreign ownership of the debt. In the early Keynesian debates about the cost of fiscal deficits, a commonly heard statement was that the debt had no cost because "we owe it to ourselves." Economists eventually recognized that that argument was wrong because servicing the debt requires distortionary taxes and because government borrowing can crowd out productive private investment. DeLong and Summers take the tax cost into account but assume away the possibility of crowding out.

Unlike in those early days, however, the majority of the U.S. government debt is now held by foreign investors, who are likely to buy an even larger share of any prospective increases in the debt. Servicing that increased debt in the future will require an increase in net exports, requiring in turn a lower real value of the dollar relative to the currencies of the United States' trading partners. Such a decline in the U.S. terms of trade reduces real incomes in the United States by raising the cost of imports in terms of forgone consumption of goods and services produced in the United States.

The second unaccounted-for cost is reduced real investment. DeLong and Summers acknowledge that under normal economic conditions an increase in the government debt crowds out productive investment in plant and equipment. They argue, however, that such crowding out is not rel-

evant now because real interest rates are low and are not increased by additional government borrowing.

But investment in plant and equipment depends on more than the rate of interest. American companies now have enormous amounts of liquid assets but are not investing those funds in new plant and equipment. Although there are many reasons for this reluctance to invest, a significant one is the concern about the future tax implications of the large and growing fiscal deficit. DeLong and Summers can argue that current deficits do not imply higher future taxes, but any business executive who reads the newspaper can anticipate that those deficits will raise future taxes on corporate profits and personal incomes.

To the extent that each dollar of government deficit reduces investment in plant and equipment, it lowers future incomes by the product of that change in capital and the real marginal product of capital. The long-run value of the real marginal product of capital is about 10 percent. Although the magnitude of the deficit-induced reduction in the capital stock is difficult to estimate, I have no doubt that it exists at the current time.

A third cost is increased economic vulnerability. A larger national debt makes the United States more vulnerable to external upward shocks to interest rates. It also makes such shocks more likely, as the recent experience in Europe demonstrates. Such interest rate increases could be caused by a lack of confidence in U.S. budget controls or by expectations of increased inflation in the United States. Here the fact noted above, that more than half of all U.S. government debt is held by foreign investors, takes on further significance.

A rise in interest rates globally because of a changing balance of supply and demand for fixed-income securities would have a greater impact on the cost of future debt service in direct proportion to the size of the national debt. Thus, any increase in the debt now could raise debt-service costs in the future in response to the higher interest rates.

More generally, the increased likelihood of higher future interest rates that results from an increased debt applies to the entire stock of the government debt as it is rolled over. It also applies to private debt and equity markets.

A final cost of increased government debt takes the form of a reduction in the government's room to maneuver. The United States may want to increase the deficit in the future for any of a variety of reasons, including countercyclical policy and national security spending. The ability to do so depends on the level of the national debt at the time. An increase in the

national debt now may reduce the government's discretion to deal with future problems.

It is well beyond the scope of this comment to evaluate the implied cost of these four adverse effects of a greater national debt, but I have no doubt that they are real and significant. Even if the comparison of the real net interest rate and the growth rate implies a low or negative cost of deficit finance at the present time, these four adverse effects imply that the benefits of fiscal policy even under the current circumstances must be substantial if they are to outweigh the total costs.

THE MULTIPLIER DeLong and Summers, relying on work by Valerie Ramey and others, estimate a multiplier on government spending of about 1.5. They note in passing that these econometric estimates of the multiplier refer to government spending and, specifically, to the national income and product accounts (NIPA) measure of government spending. They are not based on the budget measure of the fiscal policy that adds to the national debt. One reason that researchers have had a hard time estimating the fiscal multiplier is that there have been very few changes in NIPA government spending other than military spending.

DeLong and Summers's analysis is presumably intended to apply to fiscal policies like the major stimulus package enacted in 2009, the \$850 billion American Recovery and Reinvestment Act (ARRA). It is important that very little of this \$850 billion in increased government spending was counted as such in the NIPA. The major parts of the ARRA took the form of transfers to state governments and individuals as well as temporary tax cuts. The multiplier evidence that DeLong and Summers cite is therefore not relevant to this type of budget spending. Although it might be assumed that the transfers to states would lead to spending by state and local governments that could be then evaluated using spending multipliers based on NIPA data, that assumption is not warranted. It is not known to what extent the ARRA funds substituted for state governments' rainy-day funds or for the temporary tax increases (with low multiplier effects associated with temporary tax changes) that states might otherwise have enacted.

HYSTERESIS EFFECTS DeLong and Summers note that in a deep and long downturn like the one that began in 2007, workers who experience long spells of unemployment suffer a loss of human capital, and underinvestment in plant and equipment results in lost productivity. They argue that to the extent that fiscal stimulus raises GDP, it will reduce these losses of human and physical capital.

Although I accept that general point, it is unclear how substantial this effect is. To the extent that the basic multiplier on outlays like those under

ARRA is small, the counterhysteresis effects will be correspondingly small. More important, I think the authors ignore the significant countervailing effect of increased productivity that occurred in this downturn, as firms responded to the slow recovery of demand by laying off workers and changing production methods.

CONCLUSION This paper is important because it develops a benefit-cost framework for evaluating the desirability of fiscal stimulus policy by taking into account the size of the multiplier, the present value of the deadweight costs of future debt service, and the counterhysteresis effects of the improvements in human and physical capital that under certain circumstances accompany the multiplier-generated increase in GDP. The authors are careful to note that their conclusion about the desirability of a fiscal stimulus applies only to the case of a deep recession with monetary policy constrained by the zero lower bound on interest rates. My analysis within their framework, however, does not lead to the same support for the kind of fiscal policy represented by the ARRA. The reason for my skepticism about the positive benefit-cost ratio reflects different evaluations of their three components of the benefit-cost calculation. To summarize:

—I believe the multiplier evidence based on NIPA military spending cannot be applied to budget outlays that do not raise NIPA government spending.

—I think the cost of adding to the deficit includes more than just the deadweight losses associated with the additional future interest costs.

—I view the gains from counterhysteresis effects as overstated both because the multiplier effect is small and because the downturn also induced productivity-enhancing changes in production.

COMMENT BY

VALERIE A. RAMEY This paper proposes the very intriguing idea that government stimulus packages enacted during a severe downturn can be self-financing. Bradford DeLong and Lawrence Summers have actively participated in the public debate on this issue, both in and outside of government, in op-eds and on blogs, so a paper on this topic by them is of particular interest. When I was first asked to discuss the paper, I felt the same kind of anticipation that I did for the 1992 Olympics. Why? Because those were the first Olympics in which athletes who had gone professional were allowed to come back and compete with the amateurs. In the 1992 Summer Games, the U.S. basketball “Dream Team,” which included such players as Michael Jordan, Magic Johnson, and Larry Bird, beat every other team

by huge margins and won the gold. However, the returning pros in the subsequent Winter Games, which included such celebrated skaters as Katarina Witt and Brian Boitano, and Jayne Torvill and Christopher Dean, did not do as well. Although they all skated beautifully, the Olympic judges put more emphasis on precision than did the audiences they had performed for as pros. Thus, Torvill and Dean captured only the bronze and the rest did not win medals at all. The question then is, Are DeLong and Summers the “Dream Team,” or are they Torvill and Dean?

I will first summarize DeLong and Summers’s central idea and highlight some notable elements of their model. Since a key part of their hypothesis is the ability of government spending to reverse hysteresis effects, I will offer some evidence that can be viewed as a test of these effects. I will then assess the notion, which also appears elsewhere in the literature, that the fiscal multiplier may be higher when interest rates are at the zero lower bound and economic slack is high. Finally, I will offer a word of caution on extrapolating from past interest rates.

HYSTERESIS, THE ZERO LOWER BOUND, AND FISCAL STIMULUS DeLong and Summers very clearly present their main idea and their view of how the economy works. They argue that most of the time output is supply-determined and equal to potential output, as in a neoclassical model. Government spending at such times has no impact on output. During times such as the Great Recession and its aftermath, however, output is below potential and is demand-driven. In such a depressed economy, government spending can raise output.

In addition, the authors assume that current output levels can have an effect on future potential output, and thus on actual future output during supply-determined times—a hysteresis effect. Olivier Blanchard and Summers (1986) first introduced the idea of hysteresis effects in the context of lingering high unemployment in Europe, and they appealed to an insider-outsider theory of labor markets to motivate the idea. In the present context, DeLong and Summers appeal to various factors, such as the deterioration of the skills and labor force attachment of the unemployed and the long-term effect on capital of depressed investment rates. They present a very useful summary of the micro evidence on persistence of labor supply decisions as well as macro evidence on the link between financial crises and subsequent output growth.

On top of this structure, the authors also consider the unusually low interest rates prevailing in the U.S. economy today. These low rates potentially have two effects. First, as numerous papers have argued, when nominal interest rates hit their zero lower bound (ZLB), monetary policy becomes

impotent and fiscal policy can become more powerful than usual. Second, the low interest rates make budget deficits very cheap to finance.

This view of how the economy works naturally leads to DeLong and Summers's main conclusion: for a variety of parameter values, short-run increases in government spending during a slump can pay for themselves in the long run. The obvious implication is that fear of large budget deficits should not prevent the federal government from enacting another stimulus package in order to boost aggregate demand. This conclusion is essentially the Keynesian version of so-called supply-side economics, which the Reagan administration used to argue that tax cuts could stimulate the economy enough so that tax revenue would actually rise. Another way to look at it is as the new version of Say's Law: "In a depressed economy, government spending creates its own financing."

IS A SIMPLE MODEL BETTER? DeLong and Summers's model has more in common with standard undergraduate textbook macroeconomic models than with the typical models used at the graduate level or in research papers. For example, there is no discussion of their assumptions about fundamentals such as preferences, technology, and resource constraints. Also, the model is missing the "GE" of DSGE models, for there is no general equilibrium. The paper is silent on why interest rates are so low and why they should be expected to remain so low. Overall, the model is quite stripped down relative to standard modern macroeconomic models.

There are advantages and disadvantages of presenting this idea in such a simplified model. The advantages are several. First, because the model is so stark, the idea is very clearly presented and not obfuscated by inessential technical details. Second, because the idea is not embedded in one of the standard macroeconomic models, it avoids invoking those models' sometimes questionable assumptions.

However, the reason that modern macroeconomics has moved to models with carefully specified assumptions and microfoundations is that without them, one can often end up implicitly making contradictory or dubious assumptions. The reason that one can find many faults with modern macroeconomic models is precisely that they are explicit about their assumptions. Although these models have a long way to go to find better ways to model the economy, I do not think that replacing them with a model based on imprecisely specified intuitive ideas is an improvement. Indeed, I see that as a disadvantage of the approach.

THE KEY RELATIONSHIP IN THE MODEL DeLong and Summers's idea boils down to one key mathematical expression that they derive from their

assumptions about government spending multipliers and hysteresis. According to their model, an increase in government spending to stimulate the economy during a recession can be self-financing (in the sense of generating an increase in annual tax revenue sufficient to pay for the increase in annual debt service) if the following condition holds:

$$r < g + \frac{\eta\mu\tau}{1 - \mu\tau},$$

where r is the interest rate on government bonds, g is the growth rate of potential output, η is the hysteresis effect of current output on future potential output, τ is the tax rate, and μ is the government spending multiplier. The intuition behind this equation is as follows. If output is below potential, it can be spurred by an increase in government spending. The multiplier μ is the measure of how much output rises for a given dollar increase in government spending. This increase in current output is then translated into higher future potential output through the hysteresis effect η . These two effects imply that for plausible values of μ and η and a given tax rate τ , future annual tax revenue will increase more than the required increase in annual debt service. I will devote the rest of my discussion to exploring some evidence on three of the five parameters of this equation: the hysteresis parameter η , the government spending multiplier μ , and the long-term interest rate r .

THE HYSTERESIS PARAMETER DeLong and Summers include a very nice summary of the literature on various features of the economy that could lead to hysteresis. As they acknowledge, however, estimating the extent of hysteresis is very difficult, since it is difficult to distinguish the lingering effects of the recession itself (that is, state dependence) from the continuing effects of the unobserved forces that caused the recession. Nevertheless, the evidence the authors compile is quite interesting. One of the mechanisms they discuss is the potential loss of worker skills or labor force attachment that might result from extended periods of unemployment. The recent evidence presented by Steven Davis and Till von Wachter (2011) of significant and persistent losses in income to workers displaced during a recession is certainly suggestive. As Robert Hall points out in his discussion of that paper, however, whether losses to individual workers represent social losses or just redistributions of rents remains to be seen. If they are the latter, then these losses do not represent actual losses in worker skills.

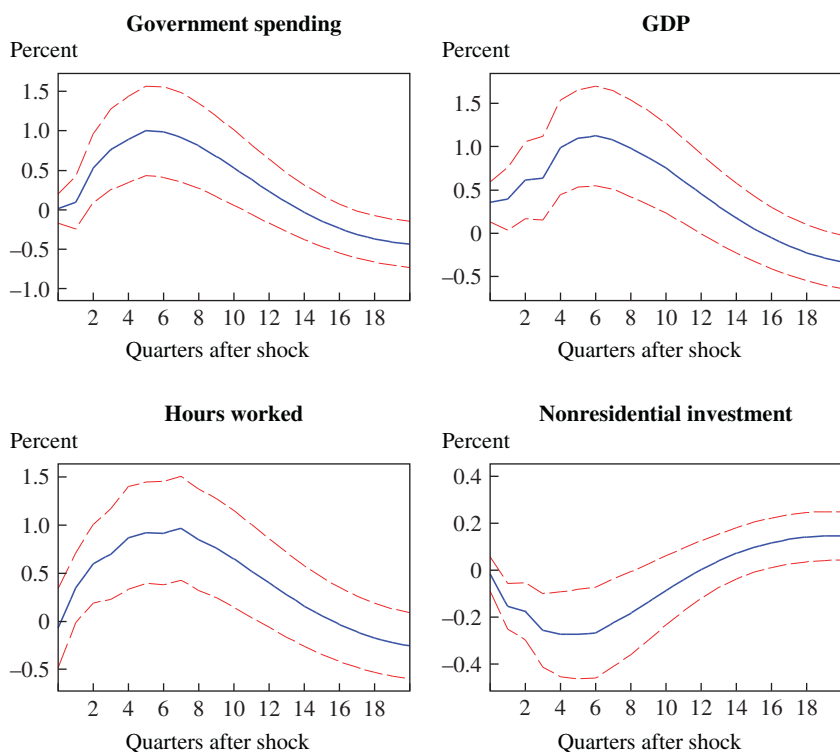
DeLong and Summers discuss a second mechanism for hysteresis that works through private investment: shortfalls in private investment during a recession, they argue, can lead to persistent effects through reductions in the capital stock. It is certainly the case that a prolonged slump in invest-

ment can lead to a significantly lower capital stock and hence lower potential GDP. As I will argue below, however, it is not clear that government spending can reverse this effect.

As DeLong and Summers themselves recognize, most of their arguments suggest persistent, but not permanent, effects. A positive depreciation rate on the hysteresis effect can have a sizable effect on their calculations. For example, consider the simulation results in their table 2, which indicate that for η of 0.025 and μ of 1.5, a stimulus package is self-financing as long as the real government interest rate is below 4.95 percent. But suppose, using the same parameters, that the hysteresis effect has a depreciation rate of 10 percent per year. Then only half of the necessary tax revenue is being collected 6 years in the future. Thus, their calculations hinge importantly on their assumption of *permanent* hysteresis effects.

For the sake of argument, suppose that hysteresis effects are indeed permanent. DeLong and Summers's argument still requires another assumption to support their policy prescription of more stimulus spending: they must assume that raising output with government spending can reverse the hysteresis effect. It is not obvious to me that an increase in government spending would create the private investment and skill-building jobs required to do that.

Even without specifying the individual mechanisms, one can test this hypothesis on U.S. data. In particular, if DeLong and Summers are correct that a change in real government spending G raises real GDP Y in the short run, it should have a persistent effect on output. That is, if one can identify exogenous movements in government spending that have led to temporary increases in real output, those increases should have a much more persistent effect on output if there are hysteresis effects. To study this, I use my analysis from Ramey (2011, 2012), which identifies exogenous shocks from military events that generated changes in the expected present discounted value of government spending. I also use a method of identification like that in Blanchard and Roberto Perotti (2002) to show that the results are not limited to just my identification method. The models are estimated from 1939Q1 to 2008Q4. Estimation is by vector autoregressions (VARs) containing log real government spending per capita, log real GDP per capita, the average marginal tax rate from Robert Barro and Charles Redlick (2011), the 3-month Treasury bill rate, log total hours worked per capita, and log real nonresidential investment per capita (all in levels). The Blanchard and Perotti structural VAR (SVAR) identifies the shock to be the shock to government spending, ordered first in the VAR. My EVAR ("expectational VAR") includes my military news variable ordered first and

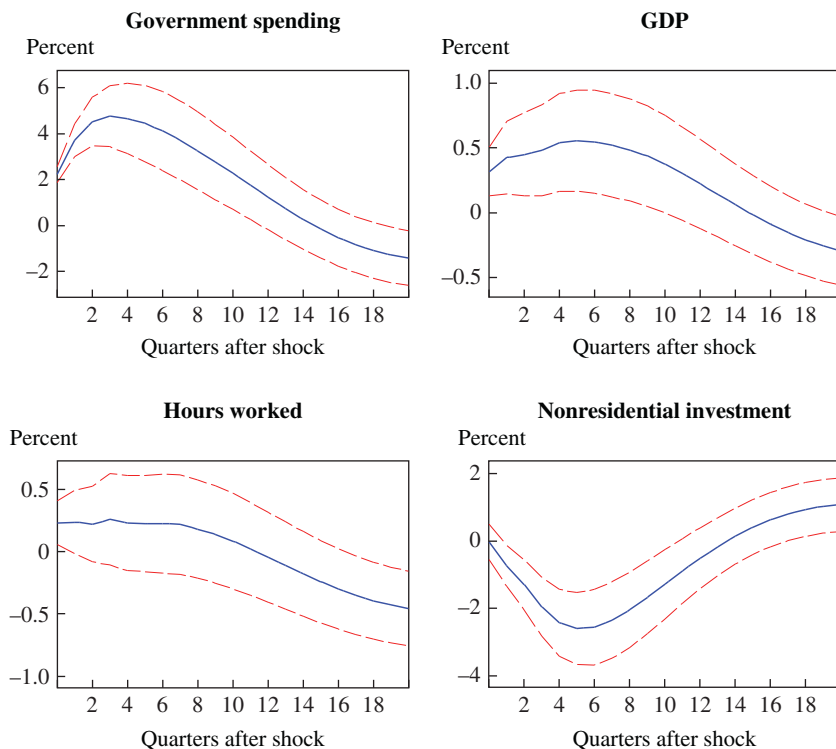
Figure 1. Impulse Response Functions from an Expectational VAR^a

Source: Author's estimates based on the expectational VAR described in the text.

a. The sample is quarterly data for the United States from 1939 through 2008. Dashed lines denote 95 percent confidence intervals based on bootstrap standard errors.

uses shocks to it as the government spending shock. Four lags are used and a quadratic trend is included.

My figures 1 and 2 show the impulse responses (with 95 percent confidence intervals based on bootstrap standard errors) of four of the variables of interest: government spending, real GDP, total hours, and nonresidential investment. In both specifications, a shock raises both government spending and real GDP. They both peak around 6 quarters after the shock and are back to normal by 16 quarters. Total hours also rise, but nonresidential investment falls. Thus, in the historical data, investment is moving in the opposite direction from that which would produce the counterhysteresis effects that the authors argue for. Nor is there evidence of a persistent effect of government spending on real GDP.

Figure 2. Impulse Response Functions from a Structural VAR^a

Source: Author's estimates based on the structural VAR of Blanchard and Perotti (2002) described in the text.

a. The sample is quarterly data for the United States from 1939 through 2008. Dashed lines denote 95 percent confidence intervals based on bootstrap standard errors.

Since my news variable captures only movements in government spending based on military events, one might wonder whether other types of government spending would have more long-lasting effects. The results using Blanchard and Perotti's framework use a shock to all types of government spending, and yet there is no more evidence of persistent effects on output. In sum, this evidence does not support the notion that an increase in government spending that raises output in the short run has lingering effects on output.

THE GOVERNMENT SPENDING MULTIPLIER A recent paper of mine (Ramey 2012) uses a more precise way to estimate the multiplier in both a VAR and an instrumental variables regression. In particular, it looks at the effect of government spending on private spending ($Y - G$). This method

indicates that the multiplier is significantly below unity—about 0.5 when tax effects are accounted for. DeLong and Summers and numerous others have argued, however, that the multiplier may be higher when there is slack in the economy (Auerbach and Gorodnichenko forthcoming) or when interest rates are at the zero lower bound (Eggertsson 2001).

In principle, it is possible to test this hypothesis on historical data. In work in progress, I have been studying the period from 1933 to 1951. As my figure 3 shows, this period was characterized by very low interest rates, similar to today's, as well as very high unemployment rates for much of the period. Of course, the presence of World War II, with patriotism raising labor force participation rates and controls on the economy dampening consumer spending, make interpretation of the period very complex. It is nonetheless interesting to at least search for differential multipliers during this period.

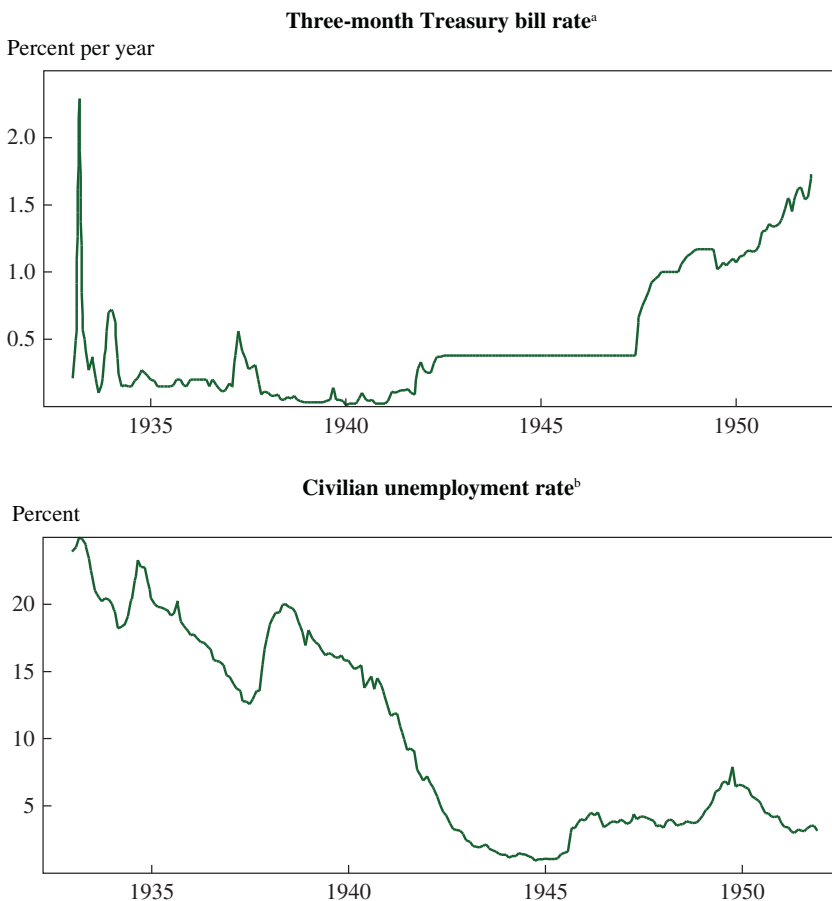
For this period, I thus estimate the following equation:

$$\frac{\Delta Y_t}{Y_{t-1}} = \beta_0 + \beta_1 \frac{\Delta G_t}{Y_{t-1}} + \beta_2 \frac{\Delta Y_{t-1}}{Y_{t-2}} + I_t \left[\beta'_0 + \beta'_1 \frac{\Delta G_t}{Y_{t-1}} + \beta'_2 \frac{\Delta Y_{t-1}}{Y_{t-2}} \right] + \varepsilon_t,$$

where I is a dummy variable that takes the value of 1 when the unemployment rate is above 7 percent and zero otherwise, and ε is the error term. I allow all of the coefficients, including the multipliers, to vary according to whether the unemployment rate is above or below 7 percent. The coefficient β_1 gives the multiplier when the unemployment rate is below 7 percent, and $\beta_1 + \beta'_1$ gives the multiplier when the unemployment rate is above 7 percent.

The data used are monthly from January 1933 to March 1951 (the month when the Treasury Accord restoring Federal Reserve independence was signed). The GDP and government spending data are from Robert Gordon and Robert Krenn (2010). The unemployment series are based on my data collection and include emergency workers. To address both the possible endogeneity of government spending and the impact of measurement error from the way that Gordon and Krenn construct their interpolated series, I instrument for government spending growth with lags 2 through 4 of government spending growth (relative to GDP).

My estimate of the multiplier β_1 when the unemployment rate is below 7 percent is 0.581, with a standard error of 0.119. The increment to the multiplier during slack times is estimated to be -0.012 , with a standard error of 0.535. Thus, the regression provides no evidence of a higher multiplier during slack times, although the high standard error indicates substantial uncertainty.

Figure 3. Short-Term Interest Rates and the Unemployment Rate, United States, 1933–51

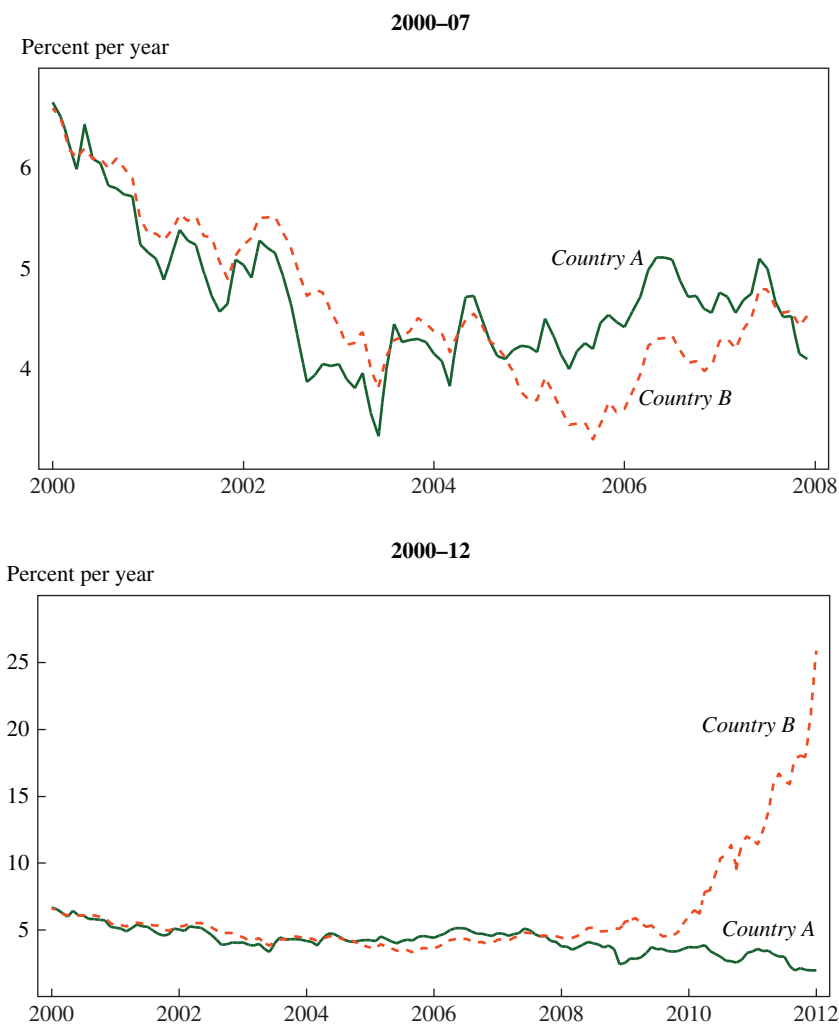
Sources: NBER Macrohistory Database, the Conference Board, *Survey of Current Business*, and author's calculations.

a. NBER Macrohistory Database series 13029b. Data are originally from the Federal Reserve Board.

b. Constructed from monthly series on the numbers of unemployed (including emergency workers) and employed workers. Underlying data from the NBER Macrohistory Database, the Conference Board, and the *Survey of Current Business* were seasonally adjusted, scaled to match the annual unemployment series compiled by David R. Weir, University of Michigan, and then spliced.

This simple analysis finds no support for multipliers that are higher during times of slack and accommodative monetary policies. The analysis presented above is quite simple, and there is some evidence to the contrary in later periods, so more research should be done on this issue.

THE GOVERNMENT INTEREST RATE DeLong and Summers look at historical data on long-term government interest rates to argue that it is unlikely

Figure 4. Long-Term Interest Rates on Government Debt since 2000 in Two Countries

Source: Organisation for Economic Co-operation and Development, Main Economic Indicators.

that interest rates will rise significantly. Here it is wise to bear in mind the Lucas critique (Lucas 1976). In particular, the historical data the authors examine were not generated in a regime in which entitlements were projected to lead to ever-rising deficits as in the current situation.

To illustrate the perils of extrapolating from the past, consider the “tale of two countries” told in my figure 4. The top panel shows interest rates on

long-term government bonds for two countries from 2000 through 2007. Both countries displayed similar patterns in interest rates over this period.

Now consider the paths of interest rates in the two countries when the sample is extended through 2011 (bottom panel). The interest rate for country A remains low while that for country B suddenly explodes and reaches 25 percent. Country A is the United States, and country B is Greece. Simply extrapolating from the past behavior of interest rates would never have led one to predict that interest rates would rise so far in Greece. Thus, current low interest rates should not be taken as a sign of future low interest rates.

CONCLUSIONS DeLong and Summers present the very intriguing idea that government spending can be self-financing when used to stimulate an economy in which output is below potential. Although I have concerns about the lack of rigor of their theoretical model, the idea is still quite interesting. My simple empirical investigations of the hysteresis effect and the government spending multiplier, however, indicate that those two parameters might not be as high as they need to be for this idea to work. Moreover, I have suggested caution in using current low interest rates to forecast the future path of interest rates. Nevertheless, DeLong and Summers have introduced an important new idea that clearly merits future research. This is what we expect from Olympic gold medal winners.

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GENERAL DISCUSSION Robert Hall observed that a better title for the paper would be "Eta," since the paper's surprising results all stem from the authors' beliefs about the value of their hysteresis parameter η . The other parameter values the authors used for their simulations seemed mostly reasonable and uncontroversial to Hall. He noted that although Valerie Ramey had estimated a relatively low value for the multiplier on fiscal spending, the standard error on her estimate was large and did not rule out the possibility that the authors' baseline value of 1.5 was correct. Hall also observed that some alternative ways of analyzing government spending data from World War II generated higher estimates of the multiplier. He found the authors' value for the growth rate reasonable, and although he shared Ramey's concern about the authors' real interest rate assumptions, he thought their baseline value might be reasonable as well.

For the most important parameter, η , however, Hall felt that much more work was needed to arrive at a credible estimate. He noted that for the interesting cases in the authors' analysis, $r - g$ is small, which makes the present value of extra output due to avoided hysteresis significant for decades into the future. In such cases, then, the appropriate value for η would be an average not just over the next decade but over many decades.

Econometrics, however, simply cannot answer the question of whether hysteresis effects, or the effects of avoided hysteresis, are significant far into the future, Hall argued. The "unit root" literature of the late 1980s had found that it was impossible to precisely estimate the persistence of shocks to GDP, yet small differences in the persistence of government spending shocks had very different implications for the analysis. And even if it were possible to estimate the long-run effects of such a shock, the United States has not experienced a government purchases shock in many years. The 2009 stimulus package did not constitute such a shock, as the positive effect of the package on government purchases was slightly more than offset by negative effects from other sources. Hall was skeptical of the suggestion of

using CBO revisions to GDP estimates as a source for estimating η , since these revisions were not made in response to fiscal shocks.

Finally, Hall thought that in their discussion of η , the authors had written with great enthusiasm about the forces that might make η strongly positive while neglecting forces that might make it less positive or negative.

Eric Swanson thought the authors were correct to use the long-term interest rate at which businesses and households can borrow as the relevant one for their analysis of the fiscal multiplier. However, he felt that the authors' view of monetary policy was too narrow in that it equated the stance of monetary policy with the current level of the federal funds rate. In his view a better measure was a medium-term interest rate such as the two-year Treasury yield, which reflects not just where the federal funds rate is at present, but also where it is expected to go over the next several quarters. This view was shaped by discussions Swanson had had with Brian Sack and research he had conducted with Sack and Refet Gürkaynak, in which they found that Federal Open Market Committee statements had large effects on the yield curve above and beyond the direct effect of changes in the federal funds rate. These effects seemed to be driven by changes in financial market expectations about the future path of the funds rate in response to forward-looking language in the FOMC statement. Thus, the FOMC appeared to have the ability to directly affect medium-term interest rates such as the two-year Treasury yield through its statements, consistent with the Eggertsson-Woodford view of monetary policy.

Taking the correct view of monetary policy mattered, Swanson continued, because if it was true that a better measure of monetary policy is the two-year Treasury yield, it was unclear that the zero lower bound was a meaningful constraint on monetary policy in 2008, 2009, and much of 2010, since yields at that maturity were consistently between 80 and 100 basis points during that period. Away from the zero lower bound, government spending can crowd out private investment by raising interest rates. Indeed, in a recent working paper with John Williams, Swanson had found that between 2008 and 2010, two-year Treasury yields were just as sensitive to economic announcements as they had been in the 1990s, when the zero bound was clearly not a constraint. Thus, the fiscal multiplier was likely to have been no larger than normal during this period. By late 2011, two-year Treasury yields had fallen below 30 basis points and were then, according to the same working paper, about half as sensitive to economic news as in the 1990s, suggesting that crowding-out effects were half as important as usual. Even in late 2011, however, corporate yields remained as sensitive to economic news as in earlier years, suggesting that crowding out

might be as important as ever. What appeared to be driving this effect was that until very recently, the private sector appeared resolute in its belief that a recovery or inflation or both would occur within the next few quarters, forcing the Federal Reserve to raise interest rates. The most recent FOMC commitments to keeping rates low through mid-2013 and late 2014 appeared to have finally altered those beliefs.

Arvind Krishnamurthy, citing work he had done with Annette Vissing-Jorgensen, thought one reason Treasury yields were currently so low was that a global shortage of safe and liquid assets had led investors to place a high safety and liquidity premium on Treasury bonds. Krishnamurthy warned that this premium could easily disappear in the next few years, which made it risky for the government to finance significant new spending with short-term bonds. Financing with long-term bonds would avoid this risk but would command higher interest rates, reflecting market expectations that the safety and liquidity premium on Treasury bonds would disappear over time. These higher rates, he thought, were the ones the authors ought to consider in their analysis. Pointing to his work with Vissing-Jorgensen on estimates of the elasticity of interest rates to the size of the government's debt, Krishnamurthy suggested that the authors should account for the likelihood that increasing government debt would push up interest rates.

Justin Wolfers endorsed Hall's view that η was the paper's key parameter. He noted that the economics profession knows amazingly little about the degree and extent of hysteresis and that exploring the policy implications of this uncertainty was itself an interesting exercise. He thought that the nonzero probability of η being positive made Summers and DeLong's analysis worthwhile. Today's high long-term unemployment rate made Wolfers worry that η was indeed positive, although he felt the paper should equally consider forces that might make η low or negative, including those identified by Ramey.

Wolfers also suggested that uncertainty about η had equally large implications to explore for monetary policy. Economists have changed their thinking about monetary policy after realizing that inflation can cast a long shadow as it shifts markets' long-run inflation expectations, and the potentially long shadow of unemployment and other forms of hysteresis could be similarly important.

Martin Baily hoped the authors would do more to tie their analysis to the current situation in the United States. Specifically, he wondered whether their model implied that the federal government ought to undertake more stimulus now, even though politics made that possibility extremely unlikely. Baily had been worried that more spending would push the nation over a

fiscal cliff, and he was therefore intrigued by DeLong and Summers's suggestion that more spending today could actually reduce future deficits.

Baily also said that he agreed that the stimulus had improved the economy relative to the outcome with no stimulus, but it was surprisingly hard to show that the 2009 stimulus bill had been instrumental in turning the economy around from a decline in annualized GDP growth of 8.9 percent in the fourth quarter of 2008 to almost 4 percent positive growth in the fourth quarter of 2009. The automatic stabilizers, he thought, might have played at least as large a role in the turnaround, and growing exports and inventories had also contributed substantially. Meanwhile government purchases had not increased much over that period, and disposable income, which the stimulus bill had explicitly sought to boost, had continued to fall.

Ricardo Reis agreed strongly with Hall and Wolfers that this was a paper about η . Most macroeconomists who had studied the unit root literature of the late 1980s equated that root with technology shocks. But Reis saw little to no evidence that the sole determinant of long-run GDP was technology. He viewed η , then, as a parameter that could capture the effects of various short-run interventions on long-run output, where different types of interventions have different values of η , and the higher its η , the larger an intervention's impact. He thought the authors should do more to explain their view that government spending, in general, has a high η , and that they would do well to differentiate among the η s of different types of government spending, such as tax rebates, infrastructure spending, or investment in science research.

Reis also agreed with Ramey that the authors ought to develop a model of what determines η , although he was indifferent as to whether this model was a dynamic stochastic general equilibrium model or something else. He noted that in recent years Philippe Aghion had made some progress in this area, modeling the mechanism through which recessions can induce lower research and development spending and so reduce potential output, as well as empirically measuring the long-run scars of recessions through R&D spending.

Christopher Carroll agreed with Hall that it was impossible to test for the existence of long-run hysteresis effects in aggregate data but was optimistic about the possibility of measuring it at the micro level. As an example, Steven Davis and Till von Wachter, in a paper in the previous issue of the *Brookings Papers*, had found that individuals laid off during recessions suffer long-run earnings losses over 50 percent larger than those experienced by individuals laid off during expansions. Carroll found this result

suggestive of important long-run hysteresis effects. Hall disagreed, however, saying his view was that the result was driven by selection effects.

Carroll also singled out a result from the Stock and Watson paper, presented before the Panel the previous day, that he found striking: One of the major reasons the recent recession appeared different from previous ones was that the economy effectively suffered a major negative monetary policy shock when interest rates hit the zero lower bound. Because of this practical barrier to negative interest rates, the federal funds rate remained stuck at a value that was two or three percentage points too high. Once one accounted for this shock, the macrodynamics of the economy were similar to those of previous recessions.

This result led Carroll to wonder whether automatic fiscal stabilizers had played a more important role than previously understood in minimizing the long-term damage from economic downturns, and if so, whether in light of the recent experience it would make sense to put even more of these stabilizers in place, to further reduce the odds of hitting the zero bound. Additional automatic stabilizers might also help circumvent the political challenges of passing one-time stimulus bills in a timely manner.

Olivier Blanchard agreed with Hall that the Congressional Budget Office's revisions to its economic forecasts could shed no light on hysteresis. In fact, all these revisions meant was that the CBO had discovered that the economy had experienced supply shocks, which affect long-run GDP growth, not that transitory shocks were having permanent effects. Blanchard also thought that pure time-series econometric techniques could not, on their own, be used to test for the existence of hysteresis, since they cannot distinguish between the effects of permanent shocks, such as a permanent increase in the price of oil, and the permanent effects of a transitory shock, such as an increase in consumption due to rising animal spirits. Finally, Blanchard thought it was a misnomer to call η a "hysteresis" parameter, since hysteresis implies a permanent effect, whereas the authors' analysis would still hold if the effects of η were long-lasting but not permanent.

Betsey Stevenson concurred with Reis that different types of government spending could have very different effects and that it was important to distinguish among them. Building on Carroll's point about automatic stabilizers, she wondered what types of stabilizers might be most desirable. For example, should education spending rise automatically when unemployment rises? Or would it be a better idea to spend more on highways?

Michael Woodford agreed with Blanchard that it was impossible to isolate evidence of hysteresis in revisions to potential output during reces-

sions. He saw the problem as fundamentally an issue of signal extraction: Even if short-run deviations from potential output had no effect on long-run potential output, one would expect to find a correlation between short-run deviations below full output and lower long-run potential output. And even if deviations from potential output really did cause long-run potential output to fall, government purchases during a recession would not necessarily prevent that. The channel through which deviations from potential output decrease long-run output might, for example, be decreased investment, in which case more government purchases would not improve long-run potential output; indeed, if these purchases negatively impact private investment, they might lower potential output. Finally, Woodford thought the authors were wrong to assume that hysteresis effects, if they exist at all, are permanent. He encouraged the authors to test the sensitivity of their results to different lengths of persistence of hysteresis.

David Romer agreed with those panelists who thought the authors should develop a more fully specified model, and he cited an example of an aspect of the economy that their model ignored that could affect their results. The authors had not specified what determines the rate of inflation. Suppose that it is determined by an accelerationist Phillips curve. In that case, higher output today would lead to higher inflation in the long run unless the Federal Reserve restricted output at some point in the future, and that, Romer suspected, would undermine the authors' result that fiscal spending at the zero lower bound has long-run benefits. Romer doubted that, in fact, an accelerationist Phillips curve characterized the U.S. economy today, but the example indicated that it would be prudent for the authors to lay out a fuller set of assumptions about the structure of the economy. Without a more complete model, readers are left to guess about the conditions under which the paper's results apply.

Martin Feldstein agreed with the authors that some aspects of cyclical weakness could have a negative influence on long-run potential output. For example, workers' skills might decay when they are out of work, and business productivity might be harmed when their capital is underutilized. However, he thought it important also to consider aspects of cyclical weakness that might work to *increase* long-run potential output. In the current economy, for example, one reason for the persistently low labor force participation rate was that many younger people were staying in school and presumably building human capital. There was also evidence that many businesses, after realizing the downturn would last a long time, had taken the opportunity to figure out how to produce more with fewer workers, which could translate into long-run productivity gains.

Responding to the discussion, Lawrence Summers said he was well aware of the costs of long-run budget deficits that Feldstein had stressed in his formal comment, and he underlined that the key to his and DeLong's argument was that, if their model is correct, short-run fiscal expansion actually reduces, not increases, the long-run deficit. He thought that investors in southern European countries appeared to be placing real stock in such a possibility today: concerns about long-run growth and competitiveness, which could be improved by fiscal expansion—and not concerns about fiscal profligacy—seemed increasingly to be a factor driving up interest rates on government bonds in the region. The fact that Spain had a smaller budget deficit than Germany just four years ago also lent support to the view that the prospect of weaker long-run growth, not deficits, was weighing most heavily on the minds of investors these days.

Summers also clarified that he and DeLong had not meant to argue that, so long as the interest rate is less than the growth rate, government ought to spend limitlessly—a view that Summers saw as akin to the fallacy that since stocks, on average, return more than bonds, one ought to try to borrow without limit to buy stocks. Rather, he and DeLong had tried to demonstrate that a combination of hysteresis and multiplier effects made government spending look attractive under economic conditions like those prevailing today. Responding to Krishnamurthy, Summers said that he and DeLong had not intended to suggest a permanence to the liquidity premium on government bonds as grounds for fiscal expansion.

Summers said that he and DeLong would certainly study Ramey's work on estimating the fiscal multiplier. However, he questioned how well the multiplier could be identified by examining the World War II period, since so much was going on in the economy alongside the burst of government spending at that time.

Summers agreed with Swanson that the federal funds rate is not a sufficient statistic for monetary policy and that, in theory, if the Federal Reserve responds to expansionary fiscal policy by doing less quantitative easing or less forward-looking signaling, it will offset expansionary policy much as it does in normal times. Summers thought that he and DeLong should consider the evidence Swanson had referenced suggesting that monetary offsetting of fiscal expansion was a real concern.

Summers agreed with the panelists who had suggested that the parameter η was of central importance to the paper. Responding to Reis and Stevenson, he did not doubt that the marginal productivities of, for example, investing in infrastructure and investing in the National Science Foundation differ in interesting and important ways, but he thought it beyond the scope of the

paper to compare their long-run supply-side effects. The real impetus for the paper, Summers said, was to explore the implications of loosening an assumption that had become popular following the unit root debate, which was that only technology shocks have permanent effects on long-run potential output. An alternative view, that reductions in aggregate demand could also have protracted effects, seemed to have substantial enough implications for optimal fiscal policy that it was worth modeling more carefully.

On whether it made sense to use revisions to CBO forecasts to test for hysteresis, Summers thought that if one added the identifying assumption that fluctuations in output between 2007 and 2010 were driven by aggregate demand shocks and not aggregate supply shocks, one could then develop an estimate of the hysteresis parameter using the CBO revisions. One might develop such an identifying assumption by more carefully specifying resource strengths.

Summers thought it reasonable to question the permanence of hysteresis effects and appreciated the skepticism that panelists had expressed about results that hinged on the weight of outcomes 75 years in the future. In fact, he and DeLong had included a decay rate on the effects of η in earlier drafts of the paper and might in the final draft either reinstate it or include an examination of debt-to-income ratios after 15 years.

Finally, Summers disagreed with Feldstein's view that current deficits could significantly reduce private investment by making investors worry about higher future taxes on corporate profits and personal income. The value of new capital investment is closely related to the value of existing capital, and the strong growth in market value of existing capital over the past two years made it seem implausible to Summers that concerns about tax hikes were reducing investors' expectations about the future profitability of capital.

